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Federated Learning for IoT: Applications, Trends, Taxonomy, Challenges, Current Solutions, and Future Directions

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ABSTRACT The rapid advancement of Internet of Things (IoT) technology has transformed the digital landscape, enabling unprecedented connectivity between devices, people, and services. Traditionally, IoT-generated data was processed through centralized, cloud-based machine learning (ML) systems, raising significant privacy, security, and network bandwidth concerns. Federated Learning (FL) presents a viable alternative by transmitting only model parameters while preserving local data privacy. Despite the growing body of research, there remains a gap in comprehensive studies on FL-enabled IoT systems. This review provides an in-depth examination of the integration of FL with IoT, highlighting how FL enhances the efficiency, robustness, and adaptability of IoT systems. The paper introduces the foundational principles of FL, followed by an exploration of its key benefits in decentralized IoT applications. It presents a comparative analysis of FL-IoT architectures using quantitative metrics and proposes a taxonomy that clarifies the complexities and variations in FL-enabled IoT systems. The challenges of deploying FL in IoT environments are discussed, along with current trends and solutions aimed at overcoming these hurdles. Furthermore, the review explores the integration of FL with emerging technologies, including foundational models (FMs), green and sustainable 6th-generation (6G) IoT networks, and deep reinforcement learning (DRL), emphasizing their role in enhancing FL's efficiency and resilience. It also covers FL frameworks and benchmarks, providing a valuable resource for researchers and practitioners in the field The article concludes by identifying promising research directions that are expected to drive future advancements in this dynamic and expanding field.

INDEX TERMS Federated learning, intelligent IoT applications, FL-architecture, foundational models, green and sustainability 6G IoT.

I. INTRODUCTION

THE INTERNET of Things (IoT) is revolutionizing our interaction with the world around us. This technology has penetrated various sectors, including smart homes and industrial automation, elevating the efficiency and convenience of our daily routines. As IoT becomes more integrated into our lives, it generates a massive volume of data that offers detailed insights into user behavior, preferences, and emerging trends [1], [2]. This surge of data from IoT devices, while beneficial, introduces several challenges. Among the concerns is the sensitive nature of the information collected, which ranges from personal user details to proprietary business data and even state-level critical information [3], [4]. Traditionally, the analysis and processing of this data rely heavily on cloud-based centralized systems, which utilize ML algorithms hosted on cloud servers [5]. While these centralized systems are robust, they raise significant privacy and security issues. Centralized data repositories accumulate vast amounts of personal data, making them a target for breaches and unauthorized access. This risk has become a focal point of public concern, especially as data rights and privacy awareness grows. People are increasingly worried about the safety of their personal information in such concentrated data stores [6].

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Additionally, IoT devices' large volume of data necessitates extensive communication bandwidth, especially since this information is frequently sent to the cloud for analysis. This constant data transmission strains communication networks and incurs considerable overhead [7].

Recognizing these challenges, there is a pressing need for an alternative approach that preserves user privacy while efficiently managing the data. FL offers a promising solution to this problem. Unlike traditional methods where vast amounts of raw data are sent to central servers, FL operates on a principle where only model parameters are transmitted. This means that devices, or nodes, process the data locally and then send only the essential model updates to the central server. This approach not only ensures heightened data privacy but also significantly reduces the communication overhead, as the bulk of the data remains on the device and only concise model parameters are communicated. Hence FL has been applied in various IoT fields including healthcare [8], [9], [10], smart cities [11], [12], and autonomous systems [13], [14].

A. RELATED WORK

Recently, a number of surveys have primarily focused on reviewing FL, for instance, [22], [23], [24], [25]. However, these works primarily focus on broad topics like Client selections, models, and algorithms related to FL, without a specific connection to IoT applications. Some studies including [15], [16], [26], explore FL-enabled IoT systems, providing insights specifically tailored to IoT applications. Khan et al. [15] presented an extensive survey that outlines recent developments in FL for IoT applications, including a set of metrics for evaluation, and taxonomy of FL over IoT network, however, a detailed discussion about heterogeneity issues and current solutions that address this problem were not provided. Similarly, [27] explores the application of FL in resource-constrained IoT environments, addressing the challenges and limitations of implementing FL algorithms due to varying computational resources. However, this study lacks discussion about application cases, frameworks, and different FL-enabled IoT architectures. Also, A study that tackles communication and computation challenges in FL is presented by [17]. Some specific studies focus on one FL-enabled IoT application such as IoT-enabled healthcare [18] and smart cities [11]. The authors of [19] presented a systematic literature review of various model aggregation algorithms and their insights. Another work [28] investigates FL applications in mobile edge networks and the challenges associated with them. Alsharif et al. [20] recently proposed a classification of FL grounded in sample partitioning and network architecture, and highlighted several FL applications in 6G IoT. However, this classification lacks comprehensiveness. Quy et al. [21] focused on applying FL to 6G IIoT systems, exploring its potential in smart healthcare, transportation, smart cities, and industry. In contrast, our work offers a more comprehensive approach, covering not only FL for 6G IIoT but also a broader range of emerging technologies, FL architecture, frameworks, and applications. Table 1 highlights key findings from previous reviews on FL-based IoT and contrasts them with our study, including the criteria for inclusion and exclusion.

B. CONTRIBUTION

While numerous surveys exist, none have sufficiently explored emerging trends in FL such as foundational models (FM), 6G, and DRL, which this survey aims to cover. This gap in the literature is the primary motivation for conducting this study. The systematic review presented here takes a markedly different approach from the methodologies outlined in Section I-A. This article offers a comprehensive overview of recent developments in the field, analyzing and summarizing works published between 2017 and September 2024.

The primary contributions of this review are summarized as follows:

- A thorough analysis of state-of-the-art FL-enabled IoT techniques including main concepts, detailed use cases, and FL's benefits in IoT is presented.
- A detailed comparative analysis of various FL architectures in IoT systems, including quantitative metrics, is presented and discussed.
- A comprehensive taxonomy of FL-base IoT systems is introduced.
- Emerging technology in FL and several key research challenges with their corresponding current solutions are discussed and highlighted.
- FL open frameworks and benchmarks are highlighted and compared.
- Emerging open issues and potential future research directions are identified.

The rest of the paper is organized as follows. Section II gives a detailed overview of FL and its application scenarios. A detailed discussion of FL-enabled IoT architecture is presented in Section III. Section IV presents a comprehensive and new FL-based IoT systems taxonomy. FL challenges and their respective solutions are discussed in Section V. Section VI provides an in-depth discussion of emerging technologies in FL. FL frameworks and benchmarks are presented in Section VII. Open issues and future directions are highlighted in Section VIII. Section VI discusses FL emerging technologies such as FM, 6G, and DRL in detail. Finally, Section IX summarizes the paper and concludes. The paper organization is shown in Fig. 1.

II. FEDERATED LEARNING AND ITS APPLICATIONS IN IoT

This section is structured as follows. Section II-A reviews a brief overview of the FL concept. Section II-B presents the importance of FL in IoT Applications. Section II-C highlights some FL Application examples in IoT.

TABLE 1. Comparison of similar recent review papers.

Work	FMs	IoT	FL Architecture	6G	Taxonomy	FL Frameworks	Application Cases
[15]	✗	✓	✓	✗	✓	✓	✗
[16]	✗	✓	✗	✗	✓	✗	✗
[17]	✗	✓	✗	✗	✓	✗	✗
[18]	✗	✓	✗	✗	✓	✓	✗
[11]	✗	✓	✗	✗	✗	✗	✗
[19]	✗	✓	✗	✗	✓	✗	✗
[20]	✗	✓	✓	✓	✓	✗	✓
[21]	✗	✓	✗	✓	✗	✗	✓
This survey	✓	✓	✓	✓	✓	✓	✓

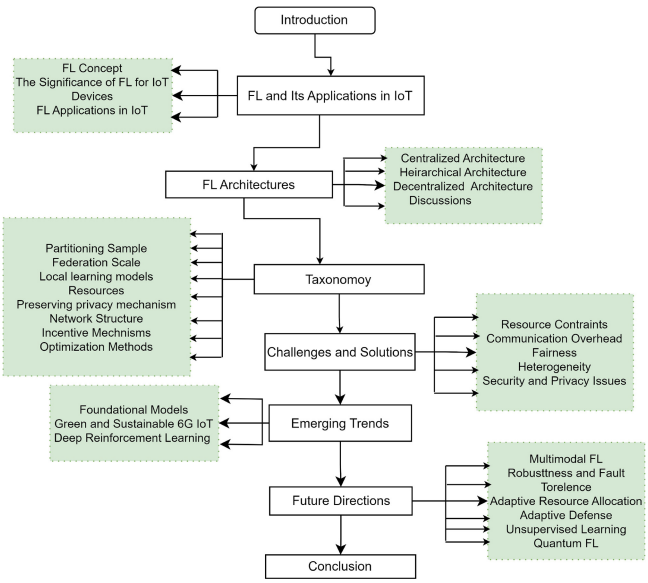


FIGURE 1. Paper organization.

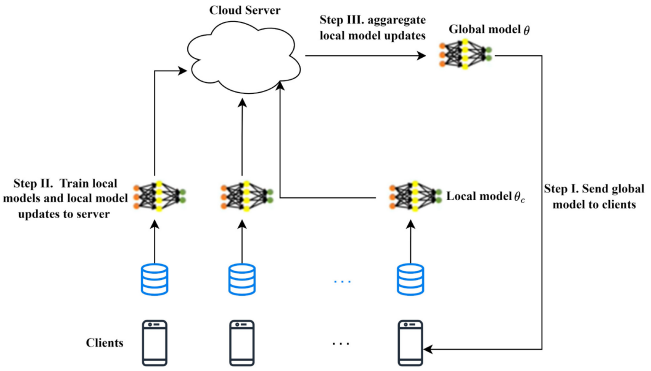


FIGURE 2. Vanilla FL architecture.

A. FL CONCEPT

The concept of FL was initially coined in 2016 by McMahan et al. [29]. It is a distributed approach in ML that allows multiple devices to collaboratively train a global model without sharing their data. This is done by sending updates to the model parameters from each device to a central server, which then aggregates the updates and sends back a new model to the devices. The system model of the Vanilla FL, also known as standard FL, setup consists of two main entities: a Central server and IoT devices as shown in Fig. 2. The server manages a network of heterogeneous IoT devices $C = |C|$, each with unique computational capabilities, and has its local dataset $D = \{D_1, \ldots, D_C\}$. The cumulative sample count across all devices is $D = \sum_{c=1}^{C} D_c$, where the following definition establishes the weighting of the local data samples:

$$\delta_c = \frac{|D_c|}{D}. \tag{1}$$

Here, δ_c represents the proportion of the c-th device's data samples relative to the total number of samples across all devices. The dataset for each device, D_c, includes several samples characterized by input-output pairs as follows: $\{(x_{i,d}^{(c)}, y_i^{(c)})\}_{i=1}^{D_c}$, where $x_{i,d}^{(c)} \in \mathbb{R}^d$ is the d-th feature of the i-th data point in the c-th dataset, and $y_i^{(c)}$ is the label for the i-th data point in the c-th dataset D_c at client $c \in C$. The Vanilla FL trains a set of local models rather than a single model as in conventional ML training. Hence, in general, the learning procedure of FL includes the following three main stages:

1) **Initialization Stage:** In this stage, before any collaborative training begins, the foundation for the FL process is laid by preparing the global model and selecting participating clients. A central server initializes the global model θ^0, which acts as the baseline for all participating clients. Then, depending on the FL configuration, a subset of clients is chosen for each training round, or all clients participate.
2) **Local Model Training Stage:** In this stage, clients receive the global model θ at round t and train it on their local data D_c, resulting in locally updated models, that is, each client trains the model on its local data. This involves forward propagation: Passing the data through the model to get predictions. Backward propagation: Adjusting the model's weights based on the prediction error. Iterating the above steps for a specified number of epochs or until convergence. This

TABLE 2. List of notations and abbreviations.

Notation	Description
C	Number of clients
D	Datasets for all clients
D_c	Dataset of c
$\{(x_{i,d}^{(c)}, y_i^{(c)})\}$	Represents the input-output pairs
θ^0	Initial global model
θ_c	Model of client c
$\nabla F_c(\theta_c, B_c)$	local gradient at the c-th client
L	Number of epochs
B	Minibatch size
$F_c(\theta_c)$	Loss function of client c
α	Learning rate
$\mathcal{L}$	Set of local iterations for each round
$F(\theta)$	Global loss function
E	Number of edges
M	Model size
AVs	Autonomous vehicles
BERT	Bidirectional Encoder Representations from Transformers
CFL	Centralized FL
CNN	Convolution Neural Network
CS	Client Selection
CV	Computer Vision
DFL	Decentralised FL
DL	Deep Learning
DNN	Deep Neural Network
DP	Differential Privacy
DRL	Deep Reinforcement Learning
FL	Federated Learning
FM	Foundational Models
FT	Fine-Tuning
GPT	Generative Pre-trained Transformer
HE	Homomorphic Encryption
HFL	Hierarchical FL
IoT	Internet of Things
KD	Knowledge Distillation
LLM	Large Language Model
ML	Machine Learning
NLP	Natural Language Processing
PT	Partial Training
SG	Stochastic Gradient
SVM	Support Vector Machine
TFP	Traffic Flow Prediction
ViT	Vision Transformer

is done by training the local model to minimize its local error as follows:

$$D_c \triangleq l\left(x_{i,d}^{(c)}, y_i^{(c)}\right) \tag{2}$$

$$f_i(\theta_c) \triangleq l(D_c; \theta_c) \tag{3}$$

This function $f_i(\theta_c)$ captures the error of the model l on the dataset D_c for each participant, and its arguments are training data and learnable parameters at round t, respectively. At each update round t, the loss function is defined as

$$F_c(\theta_c) \triangleq \frac{1}{D_c} \sum_{i \in D_c} f_i(\theta_c) \tag{4}$$

Then the local model parameters θ_c are updated as follows:

$$\theta_c = \theta_c - \alpha \nabla F_c(\theta_c) \tag{5}$$

where α is the learning rate and F_c is the local loss function of participant c. The updated local parameters θ_c and loss functions $F_c(\theta_c)$ are uploaded to the server.

3) **Model Aggregation Stage:** The central server aggregates the updates from all clients. Upon receipt of the updated local parameters θ_c and loss functions $F_c(\theta_c)$, common aggregation methods include Averaging, where the server computes the average of all received updates, or Weighted Averaging, where the updates are averaged based on the number of samples from each client. The server calculates the aggregate loss function as shown below:

$$F(\theta) \triangleq \sum_{c=1}^{C} \delta_c F_c(\theta_c). \tag{6}$$

Subsequently, the combined model parameters are derived in the following manner:

$$\theta = \sum_{c=1}^{C} \delta_c \theta_c. \tag{7}$$

Steps 2 and 3 are repeated for multiple rounds, allowing the global model to gradually improve over time. The number of rounds and the aggregation method may vary depending on the specific FL setup. The central server achieves optimal performance, marked by the minimization of the global loss function. This is typically how the FL algorithm FedAvg works as shown in Algorithm 1.

B. THE SIGNIFICANCE OF FL FOR IOT DEVICES

The decentralized nature of FL, combined with its collaborative approach and emphasis on data privacy, presents a range of notable benefits for IoT systems as listed below:

1) *Privacy and Data Security:* One of the most significant advantages of using FL in IoT is the enhancement of privacy and security of users. Traditional ML models often require data to be transferred to a central server, exposing it to potential breaches or misuse. With FL, IoT devices can train models locally, meaning the raw data never leaves the device. Only model updates or weights, which are less sensitive than raw data, are shared. This approach not only minimizes the risk of data exposure but also complies with strict data protection regulations.

Algorithm 1 FedAvg Algorithm [29]. The C Clients Are Indexed by c; B Is the Local Minibatch Size, N Is the Number of Local Epochs, and η Is the Learning Rate

Aggregation Server
Server executes:
initialize θ_0
for each round $t = 1, 2, \ldots$ **do**
  $m \leftarrow \max(C \cdot C, 1)$
  $S_t \leftarrow$ (random set of m clients)
  for each client $c \in S_t$ in parallel **do**
    $\theta_c^{t+1} \leftarrow \text{ClientUpdate}(c, \theta_t)$
  end for
  $\theta_{t+1} \leftarrow \frac{\sum_{c=1}^{C} n_c \theta_c^{t+1}}{n}$
end for

function ClientUpdate(c, θ)
  $B \leftarrow$ (split P_c into batches of size B)
  for each local epoch i from 1 to E **do**
    for batch $b \in B$ **do**
      $\theta \leftarrow \theta - \eta \nabla \ell(\theta; b)$
    end for
  end for
  return θ to server
end function

2) *Efficient Data Utilization and Bandwidth Conservation:* IoT devices, especially those operating at the edge, may have limited bandwidth and might incur significant costs when transmitting large volumes of data. FL addresses this challenge by processing data locally and only transmitting lightweight model updates to the central server. This not only reduces the amount of data transferred but also ensures efficient utilization of available bandwidth. Furthermore, by processing data on the device, real-time insights can be captured without the latency associated with transmitting data to a central server and back.
3) *Scalability and Real-time Adaptability:* The decentralized nature of FL makes it inherently scalable, a crucial feature for the ever-expanding IoT ecosystem. As more devices join the network, each can contribute to model training without overwhelming a central server. This distributed approach also allows for real-time adaptability. As each IoT device trains on its local data, the global model continuously evolves, reflecting the diverse and dynamic nature of data across devices. This ensures that the model remains relevant and accurate, catering to the specific needs and contexts of individual devices within the larger IoT framework.
4) *Collective Knowledge:* FL-enabled IoT offers another significant advantage by enabling the integration of diverse data sets from a multitude of IoT devices. This integration results in the creation of a highly detailed and accurate global model. Unlike traditional models that might rely on limited data sources, this FL-driven model benefits from a vast array of IoT devices. This diversity in data leads to enhanced learning capabilities and greater accuracy in predictions and insights, making FL a powerful tool for IoT systems.

C. FL APPLICATIONS IN IOT

Leveraging the inherent benefits of FL, a myriad of critical IoT applications have been made possible. In this section, a detailed exploration of some of these paramount applications is provided.

1) HEALTHCARE

As IoT devices increasingly integrate into daily life, safeguarding the data they collect, especially in e-health, becomes paramount. A wide range of smart wearable devices continuously monitors various health metrics of patients, encompassing heartbeat rhythms, blood pressure fluctuations, and glucose concentrations [18]. When contrasted with other data categories, health-related information stands out as exceptionally sensitive, and close connection to users' privacy concerns. Furthermore, this type of data is strictly overseen by governmental regulations and guidelines, which significantly complicates the process of sharing data without any restrictions [37]. In such a complex and highly regulated data environment, groundbreaking techniques like FL emerge as indispensable tools for the scientific community. These tools empower researchers and experts to shape cutting-edge ML models over a fragmented data landscape. One of the paramount advantages of FL is its capability to orchestrate ML training on a vast scale, spanning numerous medical institutions, all without centralizing the data.

This decentralized approach addresses the quintessential challenge of patient privacy and stringent data protection protocols. If adeptly executed, the integration of FL within the healthcare ecosystem could unlock monumental possibilities. It has the potential to galvanize the realization of precision medicine on a grand spectrum, optimizing therapeutic strategies to ensure that each patient receives the most fitting treatment precisely when they need it.

In light of this, Chen et al. [30] proposed a framework called FedHealth, utilizing federated transfer learning for smart healthcare. This approach aims to lessen divergence across domains. FedHealth employs homomorphic encryption for a secure transfer of model updates between the server and users. While it uses FedAvg for optimization, the global model might not always excel for individual users. To enhance this, transfer learning is deployed to craft personalized models. The framework's effectiveness was tested using the UCI smartphone dataset. It is worth noting that using FedProx might offer improved performance by addressing user heterogeneity. Similarly, Yuan et al. [32] introduced an FL approach for smart healthcare that uses the FL mechanism to reduce the computational burden on IoT devices during training. Their method also minimized communication overhead between the FL server and IoT devices.

TABLE 3. State art of FL-enabled healthcare.

Reference	Aim	FL Architecture	Aggregation Algorith	Local Model	Client Selection	Dataset
[30]	Create personalized healthcare models using a secure and efficient method. It combines FL and transfer learning.	Centralized.	FedAvg.	CNN	Random	UCI smartphone
[31]	Develop a method for grouped FL, where participants are organized into clusters and periodically combine their learning models.	Hierarchical	HierFedAvrg	CNN	Clustering	BIH & HAR
[32]	Reduce computation and communication burdens in healthcare IoT devices.	Centralized	Custom Algorithm	NN	Random	PhysioNet
[33]	Predicting patient mortality and hospital stay duration using electronic medical records	Centralized. Decentralized	FedAvg	K-Means	Sequential	Custom
[34]	Enhance distributed FL communication and demonstrate its effectiveness in detecting Alzheimer's disease.	Decentralized	Custom Algorithm	NN.	Random	Custom
[10]	Train models on skin lesion images using smartphones, employing transfer learning to address data scarcity in healthcare.	Decentralized.	FedAvg.	ResNet50	Random	Atlas Dermatology
[35]	Ensure medical data privacy in cross-silo exchanges using FL, highlighting the role of data mobility in advancing clinical research	Decentralized	FedAvg	NN	Random	ChestX-Ray8
[35]	Create a blockchain-enabled system for private, collaborative learning from healthcare data using diverse models.	Hierarchical	HierFedAvrg	CNN	Random	Custom

However, the framework lacks guaranteed convergence and struggles with successful learning if there is a malfunction or server failure.

Moreover, Huang et al. [33] developed an FL algorithm for predicting patient mortality and hospital stay duration using electronic medical records. This method clusters records within hospitals and uses ML models tailored to each cluster, enhancing privacy and efficiency. Another study [34] improved distributed FL communication by iterating local updates, showing its effectiveness in Alzheimer's disease detection through simulations. Authors of [10] trained models on skin lesion images using smartphones, employing transfer learning to mitigate data scarcity in healthcare. In the same direction, [35] adopted a similar approach for medical data privacy in cross-silo exchanges, emphasizing the importance of data mobility for clinical research advancements. Blockchain and smart contracts have been proposed for secure medical record exchange. Another work by [38] demonstrated the effectiveness of a decentralized CNN for Electronic Health Records with a private Interplanetary File System (IPFS), achieving results comparable to centralized models. Conversely, the study of [39] used federated distillation on a private dataset for improved performance in healthcare settings with low-quality data. Similarly, research [40] on medical image prediction for brain tumor segmentation shared local computational models among institutions, achieving over 85% accuracy with a U-Net model using private patient data. A hierarchical clustering FL for Healthcare is proposed by Zhou et al. [31]. They introduce a clustering-based method for participant selection in FL, improving results in ECG and motion monitoring with fewer participants by grouping and aggregating models. Similarly, Warnat-Herresthal et al. [35] develop a secure, blockchain-based, multiparty computation ensemble FL system that allows diverse models to collaboratively learn from healthcare data without compromising user privacy.

1) Case Study: Breast Cancer Diagnosis and Treatment: Breast cancer is one of the most common cancers affecting women worldwide, and is the second leading cause of cancer-related deaths in women worldwide [41], [42]. Early detection of aggressive breast cancer is crucial, especially for women over 40, but remains challenging, particularly with mammography [43]. AI systems can identify patterns, solve problems, and interact naturally [44]. AI improves breast cancer detection in healthcare, reducing unnecessary treatments and enhancing survival. However, challenges persist, including gaps in ML understanding among medical professionals and concerns over data privacy and ethics [45], [46].

FL offers a solution by enabling hospitals and research institutions to collaboratively train ML models for breast cancer diagnosis without sharing sensitive patient data. The process begins with each institution locally training a model using its unique datasets, which include mammograms, biopsy results, genetic information, and patient histories. These datasets are often diverse, reflecting different demographic and clinical backgrounds, making them valuable for developing a comprehensive diagnostic tool [47]. Once local models are trained, their parameters are securely transmitted to a central server where they are aggregated using FL

TABLE 4. Dataset characteristics summary [36].

Hospital	Image resolution (MP)	Detector type	# Train	# Val.	# Test
1	23.04	Direct	22,933	3,366	6,534
2	0.02 - 4.39	Direct	8,365	1,216	2,568
3	4.39 - 13.63	Direct	44,115	6,336	12,676
4	4 - 28	Direct/Scintillator	7,219	1,030	2,069
5	8.48 - 13.63	Direct	6,023	983	1,822
6	8.6 - 13.63	Direct	6,874	853	1,727
7	1 - 136	Direct/Scintillator	4,021	664	1,288

algorithms such as FedAvg. This aggregation process creates a global model that benefits from the diverse data while preserving patient privacy. The aggregated global model, now enriched with knowledge from diverse datasets, is redistributed to the participating institutions. This model can then be further fine-tuned with local data, enhancing its predictive performance and making it more applicable to the specific demographics of each institution. The process is iterative, with local models being trained, aggregated, and redistributed repeatedly until the global model converges to satisfactory accuracy and performance as discussed in Section II-A. This approach achieves accuracy comparable to centralized models while leveraging diverse datasets and maintaining patient data security and privacy.

The work by [36] serves as a case study example for breast cancer diagnosis. Classifying breast density is crucial in breast imaging, as it helps assess the amount of fibroglandular tissue, which is linked to the patient's risk of developing breast cancer [48]. Datasets of seven international hospitals have been utilized in this study. The study involved client data with varying detector types, image resolutions, mammography types, and differing numbers of training images, reflecting non-IID datasets as shown in Table 4. These variations, caused by differences in imaging protocols and mammography systems across sites, were intentionally left unstandardized to investigate the domain shift challenges associated with non-IID data. A DenseNet-121 model, extended with a fully connected layer, classifies mammography images into four categories. The FL framework, implemented in TensorFlow1 and using the NVIDIA Clara Train SDK2, manages server-client communication and standardizes training across clients using GPUs with at least 12 GB of memory.

All mammography images were normalized to [0, 1] and resized to 224 $\times$ 224 pixels. Both left and right breast images from all views were included. Each client split their dataset into training, validation, and testing sets at the patient level as shown in Table 4. Inference involved averaging predictions from all patient images for a final prediction. Clients trained for one epoch before sending their updated weights to the server for aggregation, which employed a weighted sum method. Training used an initial learning rate of 1e-4, Adam optimization, and a mini-batch size 32. Random flips, rotations, and intensity shifts were applied to prevent overfitting. FL training ran for 300 rounds, taking around 36 hours. The best local models were then shared and tested across clients for generalizability. A further experiment involved local fine-tuning with each client's best model post-FL to enhance local performance. Cohen's linear weighted kappa3 was used to assess model performance, ensuring comparability with previous studies on mammography.

While this study showcases the potential of FL in medical imaging, it faces challenges in maintaining consistent model performance across diverse data and requires high computational resources. Additionally, using only one epoch per client may limit model convergence. These issues suggest that improving training efficiency and adaptability is essential for broader application.

2) SMART CITY

FL is playing a pivotal role in the evolution of smart cities. As urban infrastructures are increasingly integrated with digital technologies, the challenge lies in effectively managing and securing vast volumes of data. Unlike traditional centralized models, FL uniquely processes data directly on the devices, improving both privacy and security. This on-device approach not only harnesses the computational power of the countless IoT devices scattered throughout a city but also facilitates instantaneous analytics. Such real-time data processing becomes essential in scenarios like traffic management or emergency responses, where immediacy can be of the essence. In that direction, Albaseer et al. [49] introduced the (FedSem) method for handling unlabeled data in smart cities using semi-supervised learning. This method first trains using only labeled data and then incorporates unlabeled data through pseudo-labeling to boost performance. The method was evaluated on the German traffic sign dataset across diverse scenarios. While it enhances data utilization and maintains privacy, FedSem struggles with highly biased training data. Beyond this, FL's methodology, which prioritizes transmitting concise model updates over raw data, streamlines communication and reduces bandwidth demands. Its distributed framework also offers resilience; even if parts of the system falter, the broader network remains operational, ensuring uninterrupted city services. This adaptability and scalability position FL as a sustainable solution, especially as cities burgeon with data. By analyzing data at its origin, FL paves the way for hyper-personalized urban services, whether it is adaptive traffic systems or dynamic public transport routes.

In the tapestry of smart city development, FL emerges as the thread weaving together innovation, efficiency, and security. A study supporting this was conducted by [50]. This study presents a privacy-focused method using FL to forecast vehicle traffic in smart cities. They employed the FedGRU, a gated recurrent unit neural network within FL, which ensures security by updating learning models through modal aggregation rather than sharing raw data. Reference [51] presented a similar work. It combined blockchain with FL for automated control in heavy-haul railway systems, enabling distributed ML without central servers. Moreover, Elbir et al. [55] explored the use of FL

TABLE 5. State art of FL-enabled smart cities.

Reference	Aim	FL Architecture	Aggregation Algorith	Local Model	Client Selection	Dataset
[49]	Handle unlabeled data in smart cities, focusing on its initial use of labeled data and subsequent integration of unlabeled data through pseudo-labeling	Centralized.	FedSem.	DNN	Random	GTSDB
[50]	Presents a privacy-focused method using FL to forecast vehicle traffic in smart cities.	Centralized	FedAvg	GRU	Clustering	PeMS
[51]	combine blockchain with FL for automated control in heavy-haul railway systems, enabling distributed ML without central servers	Decentralized	Custom Algorithm	SVM	Random	Custom
[52]	Presents method for edge caching in vehicles to enhance privacy, cut communication costs, and accommodate vehicle mobility.	Hierarchical	MPCF	AE	Sequential	MovieLens
[53]	Develop framework for elastic edge-to-cloud model aggregation from sensor data using FL.	Hierarchical	FedAvg	VGGNet	Random, full	CIFAR-10
[54]	Protect intelligent transportation systems from attacks using quantum-based FL.	Centralized.	FedAvg.	CNN	Random	MNIST

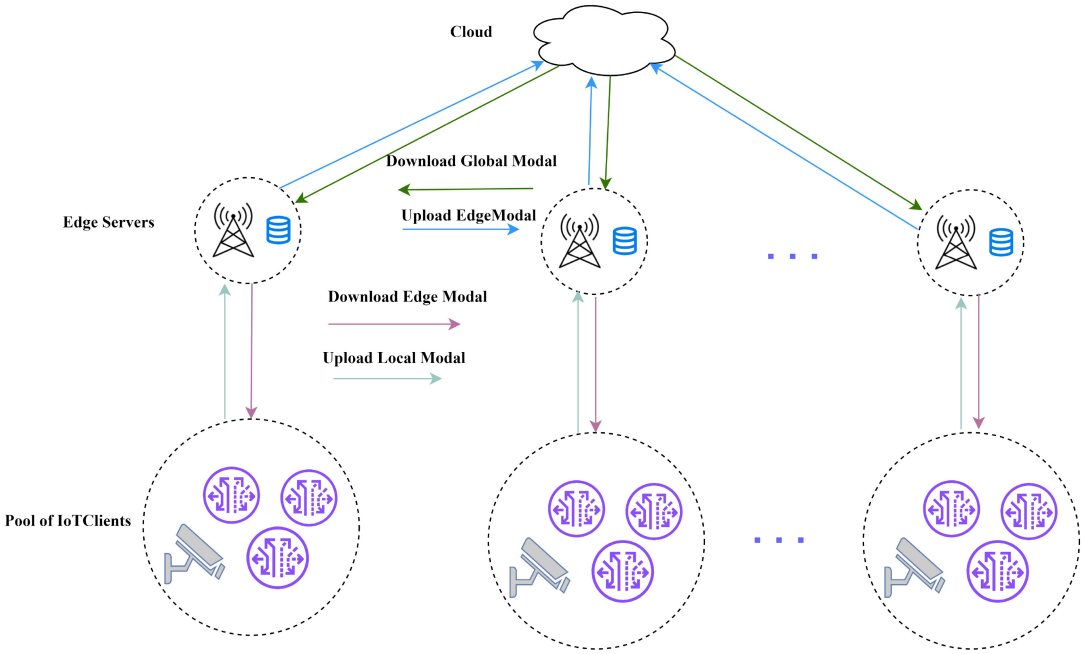


FIGURE 3. HFL framework for TFP in smart cities.

in vehicular ML applications, discussing future challenges and directions. Yu et al. [52] proposed a proactive, mobility-aware content caching strategy at network edges, adapting to vehicle mobility patterns. Another study [56] used FL to predict vehicle speeds while preserving location privacy. Also, [57] addressed operational challenges in deploying FL for smart transportation, noting the resilience of roadside FL models in fault recovery with connected vehicle intelligence. Chen et al. [53] developed a framework for elastic edge-to-cloud model aggregation from sensor data using FL. Lastly, the work of [54] introduced a quantum-based FL framework to protect intelligent transportation systems from attacks.

1) Case Study: Traffic Flow Prediction (TFP): Urban traffic congestion presents a significant challenge for modern cities, leading to increased travel times, pollution, and resource consumption [58], [59]. Accurate and timely traffic flow prediction is essential for optimizing urban traffic management and mitigating these issues [60]. FL enables different districts within a city to collaboratively train TFP models while preserving data privacy.

As shown in Fig. 3 the TFP model considers a hierarchical cloud-edge-client network. In this scenario, a smart city is divided into multiple districts, each represented as a cluster in the FL framework. Within each district, various intersections are monitored by smart cameras and sensors that serve as IoT

clients. These IoT clients collect real-time traffic data and train local models based on this data. As it can be observed TFP network comprises a CS and a set of $\mathcal{E} = \{1, 2, \ldots, E\}$ ESs that manage the communications and computations of the IoT devices represented by the set $\mathcal{C} = \{1, 2, \ldots, C\}$ CLs. Each ES e is geographically close to its subset of IoT CLs and functions both as a primary aggregator of local model updates and as a relay, channeling the aggregated updates toward the CS. In each training round t, the CS disseminates the updated global model back to the ESs. The ESs then manage the distribution and collection of model updates from and to the CLs for which they are responsible. Datasets $D_1, D_2, \ldots, D_c$ held by CLs are assumed to feature a non-independent and Identically Distributed (non-IID) and an imbalanced distribution, posing additional challenges to the learning process. The aggregate of the dataset sample counts across all clients is denoted by $D = \sum_{c=1}^{C} D_c$ where $|D_c|$ is the dataset size for client c.

The dataset for each client device, D_c, includes several samples characterized by input-output pairs as follows: $\{x_{i,d}^{(c)}, y_i^{(c)}\}_{i=1}^{D_c}$, where $x_{i,d}^{(c)} \in \mathbb{R}^d$ is the d-th feature of the i-th data point in the c-th dataset, and $y_i^{(c)}$ is the label for the i-th data point in the c-th dataset D_c at client $c \in C$. TFP Flow the HFL algorithm training method which is an extension of FedAvg. In this process, during round t, the CL calculates the L gradients based on the local dataset D_c. These local gradient parameters are then collected by the ESs, which aggregate the updated model parameters and forward them to CS. The CS uses these model parameters to update the global model, creating the version $\theta^t + 1$, which is subsequently distributed to the CLs via ESs to initiate another round of global updates. Specifically, each CL computes the gradient on a randomly selected mini-batch $B_{c,\ell}^t$ from their local dataset D_c at each local iteration ℓ of round t. The local parameter θ_c^t for the c-th client is updated as follows:

$$\theta_{c,\ell+1}^t = \theta_{c,\ell}^t - \alpha \nabla F_c\left(\theta_{c,\ell}^t, B_{c,\ell}^t\right), \ell = 0, 1, \ldots, L-1 \quad (8)$$

where the stochastic gradient (SG) can be calculated using the average over the mini-batch as follows:

$$\nabla F_c\left(\theta_{c,\ell}^{,} B_{c,\ell}^t\right) = \frac{1}{B} \sum_{x,d} \nabla F\left(\theta_{c,\ell}^t, x_d\right) \quad (9)$$

This estimation should satisfy $E[\nabla F_c(\theta_{x,\ell}^t, B_{c,\ell}^t)] = \nabla F_c(\theta_{c,\ell}^t, D_c)$, ensuring the unbiased nature of the gradients. During training, the size of each mini-batch remains constant, although its contents are refreshed with each iteration. The accumulated gradient updates over the training rounds are summarized as follows:

$$\Delta\theta_c^t \triangleq \sum_{\ell \in \mathcal{L}} \nabla F_c\left(\theta_c^c, B_c^t\right) \quad (10)$$

which infers that the local model update can also be expressed as:

$$\theta_{c,L}^t - \theta_{c,0}^t = -\alpha \Delta\theta_c^t \quad (11)$$

Subsequently, after L local updates, each ES e aggregates these model updates from all associated CLs C_e as follows:

$$\Delta\theta_e^t = \sum_{c \in C_e} \sum_{\ell \in \mathcal{L}} \nabla F_c\left(\theta_{c,\ell}^t, B_{c,\ell}^t\right) = \sum_{c \in C_e} \Delta\theta_c^t \quad (12)$$

Hence, ESs upload their aggregated parameters to the CS for global aggregation and averaging, resulting in the updated global model:

$$\begin{aligned} \theta^{t+1} &:= \theta^t - \alpha \frac{\sum_{e \in \mathcal{E}} \sum_{c \in C_e} \Delta\theta_c^t}{C} \\ &= \theta^t - \alpha \frac{\sum_{e \in \mathcal{E}} \Delta\theta_e^t}{C} \end{aligned} \quad (13)$$

Upon computation of θ^{t+1}, it is redistributed to the CLs to initiate another global iteration.

Reference [50] follows this framework by presenting FedGRU and clustering-based FedGRU algorithms which are evaluated using real-world traffic flow data from the Caltrans Performance Measurement System (PeMS) database [67]. This database contains data from over 39,000 sensors in California's freeway system, capturing individual traffic events in real-time. The first two months of 2013 are used as the training data, while the third month is the testing data to predict traffic flow at a future time interval t using past data. To assess prediction accuracy, the metrics employed include MAE, MSE, RMSE, and MAPE. These metrics quantify the differences between the observed traffic flow and the predicted values. FedGRU's model architecture is optimized using DL techniques, focusing on hyperparameter tuning for input layer size, number of hidden layers, and hidden units. Through a grid search approach, the optimal configuration for FedGRU is identified as two hidden layers, each with 50 units. The results indicate that having neither too many nor too few hidden layers is crucial for performance. The effectiveness of FedGRU is compared to GRU, SAE, LSTM, and SVM using identical simulation conditions. The evaluation shows that FedGRU achieves lower MAE than its counterparts, specifically outperforming the other models on the 5-minute TFP task. An ensemble clustering-based FedGRU is used to improve TFP, with the joint-announcement protocol reducing communication overhead by 64.10% compared to centralized models.

3) AUTONOMOUS DRIVING

FL has been harnessed to advance autonomous driving by utilizing decentralized data from various vehicles [68]. This approach improves safety and adaptability in real-time without sacrificing data privacy, allowing self-driving cars to navigate diverse conditions and evolving traffic patterns better. RL is commonly utilized in autonomous driving but can be inefficient and prolonged. Liang et al. [61] address these challenges by integrating FL with RL. They introduced a method blending FL with online transfer learning to streamline the traditionally time-intensive offline model transfers. This system enabled simultaneous learning from asynchronous agents. When tested in a simulator,

TABLE 6. State art of FL-enabled autonomous vehicles.

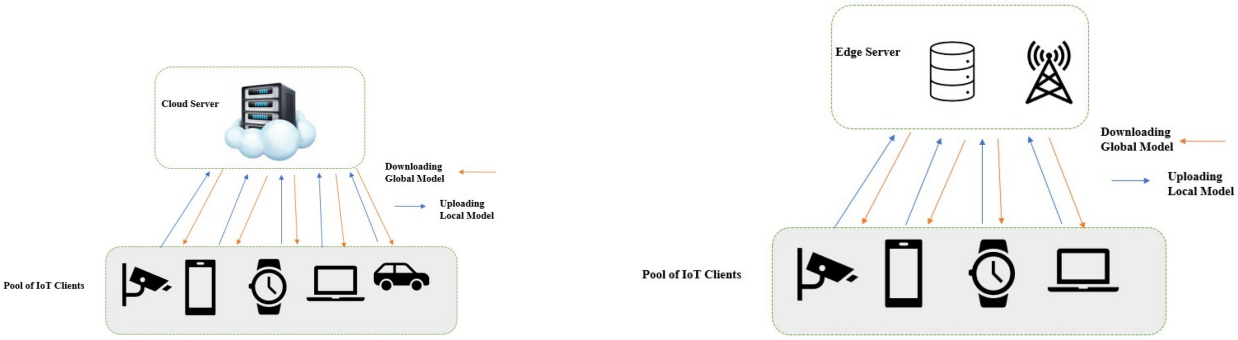
Reference	Aim	FL Architecture	Aggregation Algorith	Local Model	Client Selection	Dataset
[61]	Integrate online Federated transfer learning with RL to improve obstacle avoidance.	Centralized.	Custom FedAvg.	Pre-trained	Random	Custom
[62]	Presents FL algorithm vehicle classification to incorporate an adaptive learning rate that addresses varying accuracy demands and enhances the efficiency of the local training process.	Centralized	Fair Fed	SVM	Full	Custom
[63]	Introduce proactive caching strategy using Decentralized communications, where vehicles, aggregate the global model	Decentralized	Custom Algorithm	VAE	sequential	MovieLens 1M
[64]	Explore Decentralized FL in classifying road users and objects	Decentralized	Custom FedAvg	Pointnet	Random	nuScenes
[65]	Design a novel peer to peer FL approach, proving its resilience to byzantine faults.	Decentralized	Custom Algorithm	CNN	Schematic	MNIST, KITTI
[66]	Introduce a dynamic FL for the IoV, supported by fog federations, capable of effectively adapting to IoV landscape.	Hierarchical.	CustomFedAvg.	ResNet	Random	SUMO, GTSDB

their model outperformed traditional RL, showcasing fewer collisions and improved obstacle avoidance. Additionally, Authors of [62] introduced an adaptive algorithm leveraging FL for vehicle classification. The method streamlines training with early termination based on accuracy thresholds and uses a fairness coefficient to improve aggregation. Compared to prior methods, the proposed method offers enhanced accuracy, reduced training time, and better fairness.

Moreover, there is a type of autonomous driving called Vehicle-to-Everything (V2X). This type of autonomous driving uses a communication scheme that faces challenges like high vehicle mobility and limited node storage, which can be tackled using Distributed FL. For instance, authors of [63] introduced a proactive caching strategy using Semi-Decentralized and Peer-to-Peer (P2P) communications, where vehicles, instead of edge nodes, aggregate the global model, enhancing efficiency and autonomy. Blockchain technology further ensures data integrity and privacy. Lu et al. [69] employed deep RL to select Decentralised FL (DFL) nodes, optimizing data sharing, while another study [70] reduced system delay by considering channel dynamics. [65] developed a novel P2P DFL approach, proving its resilience to byzantine faults and comparing its performance with traditional server-based methods using MNIST and KITTI datasets. another study, [64], explored DFL in classifying road users and objects using Lidar data, achieving low latency training in a simulated realistic V2X network. Another study [71] focused on Quality of Service (QoS) in remote driving scenarios, proposing Predictive QoS (PQoS) with an RL agent to maintain optimal data transmission under low latency and reliability constraints. Lastly, authors of [72] integrated Blockchain in a fully connected vehicle network, ensuring robust privacy, system reliability, and efficient data exchange, even with variable data quality. Lastly, a different research work was done by Hammoud et al. [66] where they Introduced a dynamic FL for IoV, enhanced by fog federations using game theory for broader reach and adaptive service migration to meet changing learning needs.

1) Case Study: Object Detection in Autonomous Vehicles: Autonomous vehicles (AVs) rely heavily on real-time object detection for safe navigation [73]. These vehicles must accurately identify and classify nearby objects, such as other vehicles, pedestrians, cyclists, and obstacles, to make informed driving decisions [74], [75]. Traditional object detection models require large amounts of data collected from various sensors like LIDAR and cameras mounted on vehicles [76]. FL addresses the need for huge data, privacy concerns, bandwidth limitations, and data diversity by enabling multiple autonomous vehicles to collaboratively train a shared object detection model. Each AV collects data from its onboard LIDAR sensors and cameras. The input data consists of top-view images of the ego vehicle, capturing the surrounding environment along with LIDAR signal strengths for different elevations. The output is a classified representation of detected vehicles and objects, represented as bounding boxes around the objects obtained through preprocessing the camera images. This data varies due to differences in driving environments (urban, rural, highway), weather conditions, and sensor configurations. This leads to non-IID posing challenges for creating a robust and generalized object detection model. The AV locally trains a model using this data to perform object detection, identifying vehicles, pedestrians, and obstacles in real-time.

FL procedure for this object detection scenario follows the HFL framework discussed in the TFP case Section II-C.2.1. In this setup, AVs function as CLs, collecting local sensor data and training models. Roadside Units (RSUs) serve



(a) Cloud-Client crchitecture

(b) Edge-Client architecture

FIGURE 4. Centralized FL architecture.

as intermediaries, facilitating communication between the AVs and the system's ESs. These edge servers perform cluster-level model aggregation, while CS acts as the global aggregator, combining models from different AVs to form a robust global object detection model. An example of Object Detection in AVs is the work of [76]. KITTI [77] dataset has been used, which contains 2D color images for object detection. The dataset includes 7,481 images for training and 7,518 images for testing, covering eight distinct categories. The distribution is as follows: 910 Pedestrians, 348 Trucks, 76 Miscellaneous objects, 173 Cyclists, 293 Vans, 48 Trams, 5,632 Cars, and a single image of a Person Sitting, illustrating diverse subjects. The images are in RGB format with various dimensions. All models were implemented using the TensorFlow framework. Various scenarios were explored by adjusting the number of end devices, ranging from 10 to 100. At the global level, 100 epochs were executed, where each edge device performed 60 iterations locally. One local aggregation was then conducted between the edge and RSUs. A detailed analysis involved varying the number of local iterations and aggregation steps. The Lenet model was used with a local learning rate (η) of 0.01, a learning rate decay of 0.995, and a batch size of 32. This process was applied to both IID and non-IID data, with average accuracy and loss logged at each epoch, along with the total training time for each framework.

III. IoT-FL ARCHITECTURES

We consider horizontal FL architectures for IoT and Various architecture types exist. These IoT-FL architectures can be broadly classified into three types: Centralized FL, Hierarchical FL, and Decentralized FL.

A. CENTRALIZED FL ARCHITECTURE

The centralized architecture (Fig. 4), where a central server interacts with multiple IoT clients, is the predominant model in both practical applications and research within FL [78], [79], [80]. This framework, which involves clients independently performing gradient descent and then aggregating updates via the FedAVG algorithm after every E local ephocs, continues until reaching the desired accuracy or resource limits. The choice between cloud-based and edge-based approaches within this model depends on specific needs, with the main differences lying in communication patterns and the extent of client participation. Cloud-based approach (Fig. 4(a)) can involve participation from a large number of clients, offering a vast range of datasets which is crucial for deep learning (DL). The extensive client network in cloud-based FL contributes to a rich variety of learning patterns, making the resultant models robust and comprehensive. However, this setup faces challenges in communication speed and reliability, often due to network congestion, making the training process less efficient. Research has indicated a notable trade-off in cloud-based FL includes achieving efficient communication often requires reducing the frequency of updates at the cost of increased local computations on the client side. Hence, Various research work has been done to improve the latency and communication efficiency using data compression techniques [81], [82] as well as resource-aware methods [83].

In contrast, edge-based FL places (Fig. 4(b)) the edge server at closer proximity, like a base station, which significantly reduces latency. The computation time in this setup is comparably fast to the communication time with the edge server, facilitating a more balanced trade-off between computation and communication. This advantage makes edge-based FL preferable for scenarios requiring quick decision-making and real-time data processing. However, a limitation of this approach is the smaller number of clients each server can handle, which can lead to less diverse model updates and potentially affect the overall training effectiveness. In this direction, [84] conducted a theoretical analysis of the convergence rate in distributed gradient descent. They identified an optimal frequency for global aggregation, aiming to minimize resource consumption. Additionally, Wang et al. [85] combined DRL with FL approaches in the context of mobile-edge computing (MEC), leading to the development of an advanced edge-AI framework. This framework was specifically designed to enhance the efficiency of MEC, along with improving caching and

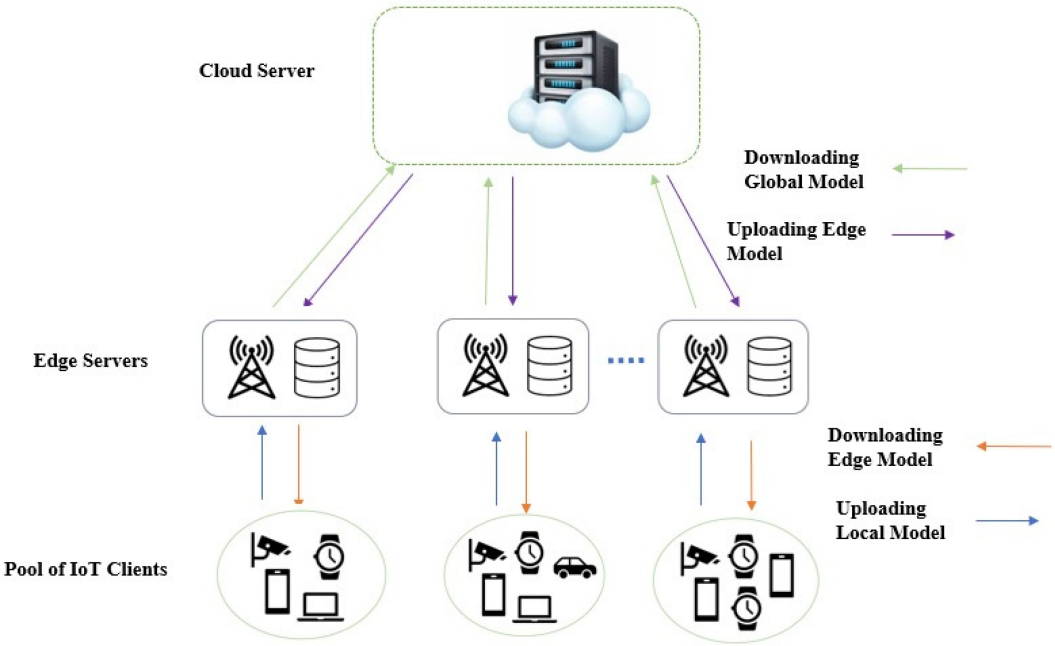


FIGURE 5. Hierarchical FL: cloud-edge-client.

communication processes. Similarly, Mills et al. [86] implemented an Adam optimization-enhanced FedAvg approach, coupled with innovative data compression techniques, to effectively decrease the number of rounds needed for convergence.

B. HIERARCHICAL FL ARCHITECTURE

In light of the discussion presented previously, it is evident that integrating a cloud server for accessing extensive training data, while simultaneously utilizing edge servers for rapid model updates with local clients, is highly beneficial. This realization has led to the development of the HFL approach, which aims to combine the strengths of both cloud-based and edge-based architectures. HFL seeks to maximize the advantages of both systems. In a HFL setup, there are two notable models: Cloud-Edge-Based and Cloud-Fog-Edge-Based FL. The Cloud-Edge-based model (Fig. 5) focuses on reducing the extensive communication demands typically associated with cloud-based FL. By integrating efficient client-edge interactions, this model significantly lowers both the runtime and the number of local iterations required, offering a more streamlined and cost-effective approach. while most studies that utilize edge servers as the global aggregator center on reducing the time to convergence, neglecting other crucial factors including the unpredictability of wireless networks [87], energy constraints at clients, and varying sizes of local data [88]. Tran et al. [89] conducted an assessment of FL over wireless networks, taking these constraints into account. In a different approach, [90] introduced a blockchain-enabled FL system, where local updates are stored on the blockchain. Meanwhile, [91] and [92] developed a HFL system spanning client, edge, and cloud levels, aiming to cut down on communication expenses compared to conventional FL methods.

On the other hand, the Cloud-Fog-Edge-Based FL model (Fig. 6) goes a step further by incorporating an additional 'fog' layer. This layer acts as an intermediary between the cloud and edge servers, further optimizing model updates. This additional layer allows the cloud server to access more data, potentially enhancing the model training beyond what is achievable with solely edge-based FL. While the potential benefits of HFL are clear, designing such a system presents several complex challenges. Extending the FAVG algorithm to a hierarchical setting raises questions about its convergence in this new context. Additionally, with two levels of model aggregation – at the edge and the cloud – determining the optimal frequency for model aggregation at each level becomes critical. Furthermore, the introduction of frequent local updates in HFL systems necessitates a careful evaluation of the latency-energy trade-off to ensure efficient and effective model training. For instance, The study by [93] integrates the Edge-Fog-Cloud architecture of FL-based framework for smart healthcare in IoT, It focuses on real-time ECG data processing in IoT medical applications, deploying and comparing the FL model across Edge, Fog, and Cloud layers for efficiency. in the same context, Saha et al. [94] proposed a greedy heuristic method that assists in selecting the optimal fog node as a global aggregator, facilitating communication between Edge and Cloud while reducing dependence on server-based execution. The FogFL architecture leverages fog nodes to reduce energy use and communication latency in resource-limited edge devices, maintaining the global model's convergence rate and boosting system reliability.

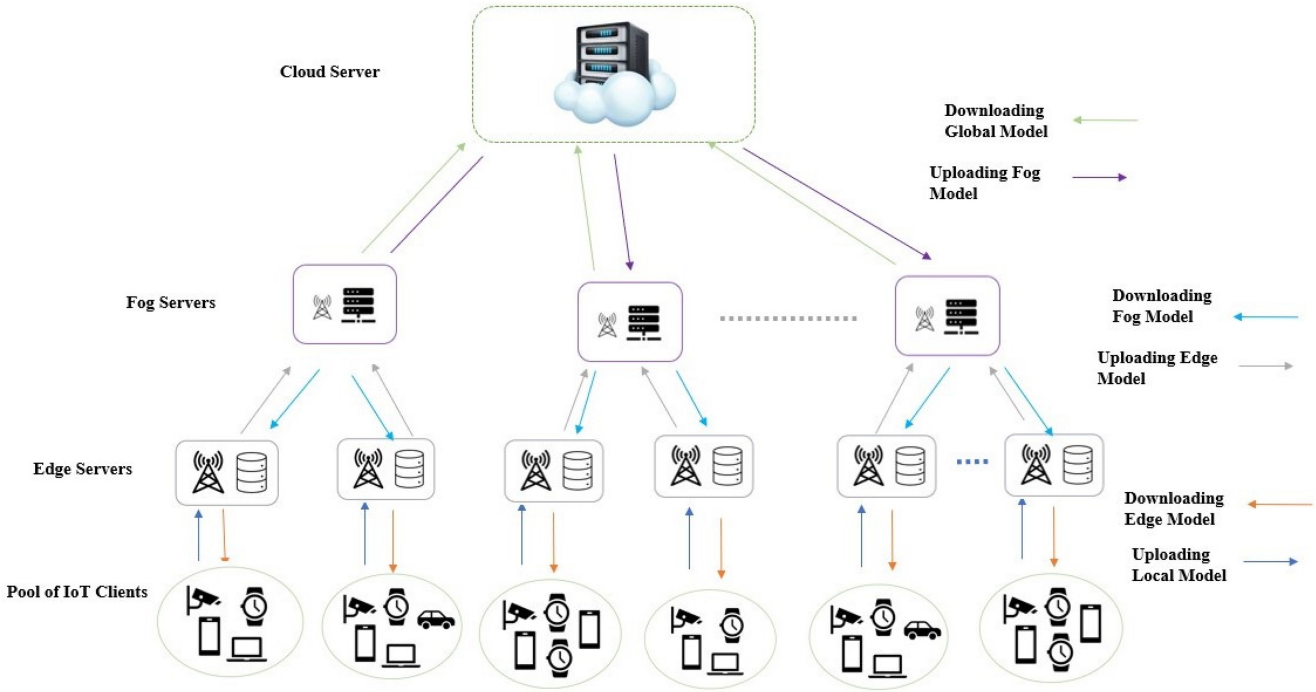


FIGURE 6. Hierarchical FL: Cloud-Fog-Edge-Client.

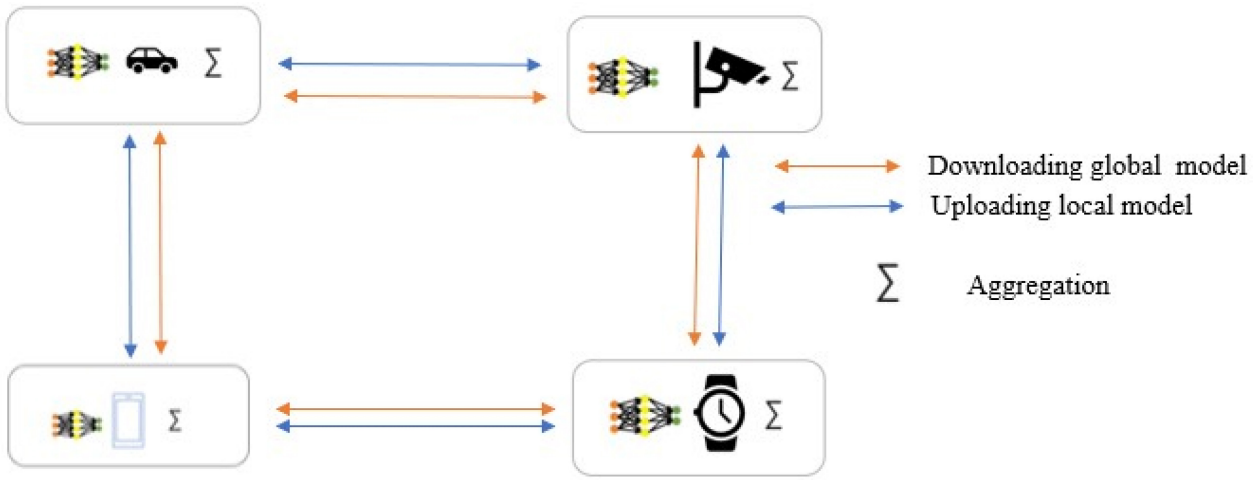


FIGURE 7. Decentralized FL architecture.

C. DECENTRALIZED FL ARCHITECTURE

Decentralized FL is a variation of the traditional FL approach, where the learning process is distributed among multiple nodes or devices, but without a central coordinating server [95]. This decentralized approach initially investigated in [100], shifts the aggregation function to the edge, distinguishing it from other alternatives discussed above. Such an architecture effectively addresses performance bottleneck issues by distributing the computational load evenly across all nodes as shown in Fig. 7. Moreover, the inherent flexibility in node connections endows the system with enhanced autonomy, allowing for swift adaptation to changes in the local environment. However, this decentralized structure introduces challenges in coordination. With each node independently managing its behavior, synchronizing collective tasks and achieving a unified global knowledge becomes complex. This often results in considerable variation in the models across different nodes, which might not be ideal for certain applications. Additionally, the decentralized model is less suitable for smaller-scale systems. Industries looking to implement small-scale decentralized systems might find them inefficient in terms of system management and overall performance, undermining their practicality in such contexts [96].

To enhance the system, incorporating advanced coordination algorithms and communication protocols could mitigate

TABLE 7. Comparison of FL architectures: DFL, CFL, and HFL.

Metric	DFL	CFL	HFL	References
Communication Times/Round	1	1	$M \times \left(\frac{C}{E} + 1\right)$	[95], [91]
Node Overhead	$M \times (C - 1)$	$M \times C$	$M \times \left(\frac{C}{E} + 1\right)$	[95], [96]
Total Transferred Data Volume per Round (MB)	$M \times (C - 1)^2$	$M \times C$	$M \times \left(\frac{C^2}{E} + C\right)$	[95]
Latency (ms per round)	$L_{DFL} =$ Comm Delay $\times (C - 1) +$ Comp Delay	$L_{CFL} =$ Comm Delay $\times C +$ Comp Delay	$L_{HFL} =$ Comm Delay $\times \left(\frac{C}{C} + 1\right) +$ Comp Delay	[91], [97]
Energy Consumption (Joules)	$E_{DFL} =$ Comm Energy $\times (C - 1) \times M +$ Comp Energy	$E_{CFL} =$ Comm Energy $\times C \times M +$ Comp Energy	$E_{HFL} =$ Comm Energy $\times \left(\frac{C^2}{E} + C\right) \times M +$ Comp Energy	[98]
Scalability (Number of Clients)	Up to 1000 clients	Up to 100 clients	Up to 500 clients	[99], [96]

the coordination challenges in a decentralized FL system. Leveraging ML techniques to predict and adapt to node behavior may also improve synchronization and model consistency. Moreover, for smaller systems, developing specialized decentralized frameworks tailored for efficiency and minimal management could make them more viable for industry applications. This tailored approach would aim to harness the benefits of decentralization, such as load balancing and adaptability while minimizing its drawbacks [101].

D. DISCUSSIONS

Following an in-depth DFL, CFL, and HFL analysis, we present a detailed quantitative comparison based on key performance metrics, including communication times per round, node overhead, latency, energy consumption, and scalability. DFL employs a peer-to-peer communication model where each node communicates once per round. The node overhead is given by $M \times (C - 1)$, where M represents the model size and C the number of clients. DFL scales well, supporting up to 1000 clients without significant performance degradation. One of its main advantages is its high fault tolerance, as the decentralized nature ensures that the system can continue operating even if several nodes fail. However, the absence of centralized aggregation can lead to slight inconsistencies in the global model due to asynchronous updates across nodes. Despite this, DFL remains a highly robust solution for large-scale systems that prioritize fault tolerance over consistency. In contrast, CFL centralizes communication, where each node communicates with the central server once per round. This results in a node overhead of $M \times C$, but scalability is limited to around 100 clients due to the central server's role as a bottleneck. While model consistency is high due to centralized aggregation, the architecture's single-point-of-failure nature and limited scalability hinder its application to larger networks. CFL is highly energy-efficient, but the lack of scalability and fault tolerance makes it less optimal for broader deployments. HFL introduces a hierarchical communication structure that mitigates the limitations of both DFL and CFL. In HFL, clients are grouped into clusters, with each cluster having a local aggregator responsible for aggregating the updates before communicating with the global server. The communication times per round are described by the formula $M \times (\frac{C}{E} + 1)$, where C is the number of clients and E is the number of clusters. This architecture allows HFL to scale up to 500 clients while reducing the communication overhead compared to DFL. Node overhead follows a similar pattern, balancing the computational load between clients and local aggregators. HFL improves fault tolerance by localizing failures within clusters, minimizing the impact of node or cluster failures on the entire system. While the model consistency in HFL is slightly lower than in CFL due to the multi-tiered aggregation, it remains higher than in DFL, making it an effective compromise between scalability, consistency, and fault tolerance.

It is worth mentioning that the architecture choices should ultimately depend on the specific application requirements, balancing trade-offs between model consistency, communication overhead, and energy consumption. The detailed comparison of these architectures is summarized in Table 7, where formulas rather than fixed numerical values are presented to allow for adaptability to various real-world settings and configurations. This flexibility ensures the metrics can be tailored to different network topologies, device capacities, and communication environments, thereby providing a robust and scalable solution for FL systems.

IV. TAXONOMY ON FL-BASED IOT SYSTEMS

Numerous studies have categorized FL within the IoT framework from various perspectives [15], [16], [102]. After examining these categorizations, we developed our own classification, as depicted in Fig. 8 Each category is explained in the following subsections.

A. PARTITIONING SAMPLE

When building a FL model, it is essential to assess the distribution of data records. This involves considering not only the unique features within the dataset but also the instances that

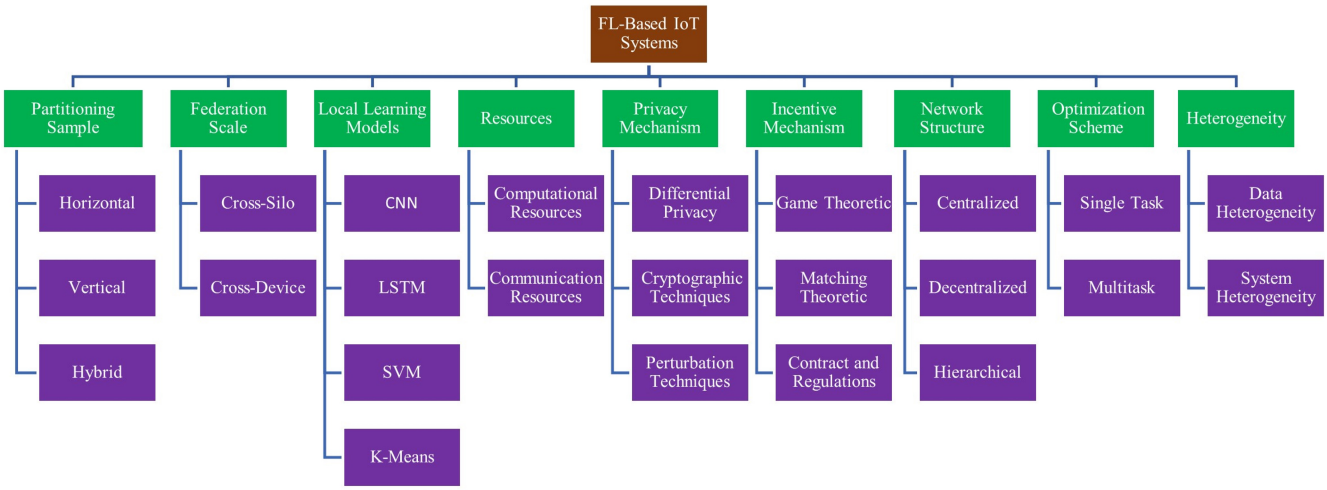


FIGURE 8. FL taxonomy.

do not intersect or share common elements which are known as the non-overlapping instances. The partitioning approach is employed to optimize model performance, strengthen security measures, enhance scalability, and ensure greater availability. FL can classified into three main categories based on data partitioning, feature space of those samples, and the nature of the application. Understanding the distinctions between these FL categories is crucial for selecting the appropriate approach based on the data distribution and privacy requirements in a given IoT application domain.

1) *Horizontal FL (HFL):* HFL involves the scenario where data is distributed horizontally across multiple parties or devices, and each party holds examples of the same feature set with different instances [103]. In other words, each party has data samples with shared attributes but differing data points. For instance, consider a healthcare association where different hospitals have patient records with the same attributes (e.g., patient age, gender, and medical history), but the patient populations are unique to each hospital. In HFL, a global model can be trained collectively without sharing specific patient information, benefiting from the collective knowledge of all hospitals. The majority of FL research adheres to the horizontal approach. In this method, clients use a common feature set to train their models, leading to the creation of a comparable global model structure.
2) *Vertical FL (VFL):* VFL addresses situations where data is distributed vertically, that is, different parties or entities hold complementary sets of attributes for the same instances [104]. This is particularly relevant when certain data attributes are sensitive and should not be shared. An example of VFL can be seen in the collaboration between a financial institution and a credit bureau. The bank may have information on customer income and transaction history, while the credit bureau possesses credit scores and payment history. By applying VFL, these two entities can collaboratively build a credit risk model without revealing the individual attributes, thus maintaining data privacy.
3) *Federated Transfer Learning (FTL):* FTL is also known as Hybrid FL and it represents a combination of Horizontal and Vertical FL, where data is distributed both horizontally and vertically among different parties or devices. it accommodates scenarios where data sharing involves a mix of shared attributes and complementary attributes [105]. For instance, consider a consortium of e-commerce platforms collaborating on a recommendation system. While they all have user behavior data (horizontal sharing), they may also hold unique user demographic information (vertical sharing), such as age and location. HFL allows them to collectively enhance their recommendation model by leveraging both shared and private data.

B. FEDERATION SCALE

In FL, two distinct categories emerge: Cross-silo FL and Cross-device FL. This classification is based on the number of client nodes and the volume of their associated data, reflecting the broader scale of the federation.

1) *Cross-device FL:* The cross-device FL paradigm is characterized by the engagement of a substantial cohort of clients operating within a homogenous domain, collectively demonstrating parallel interests toward the main global model. This approach is particularly relevant given the vast number of interconnected devices in the IoT ecosystem. It facilitates the collaborative learning process where multiple IoT devices, such as smart home appliances, wearable devices, and vehicular networks, contribute to the training of a global model [106]. Each device processes its local data and shares model updates rather than raw data, ensuring data privacy and reducing communication overhead.

This methodology is especially pertinent for scenarios where devices within a similar domain or application area aim to benefit from a shared learning experience while preserving the autonomy and security of their datasets. This type of FL utilizes resource allocation optimization techniques such as client selection [107] and device scheduling to select updates from clients that offer more valuable contributions. To reinforce active participation and ensure optimal contributions, incentive structures rooted in game theory [108] have been developed. These mechanisms are crucial in motivating clients to actively contribute to the FL process.

2) *Cross-silo FL:* Cross-silo FL addresses the challenges posed by data residing in distinct organizational or operational silos [109]. Unlike the Cross-device approach, which focuses on individual devices, Cross-silo FL emphasizes collaboration between different data repositories or silos within the IoT landscape [110]. For instance, data from various smart city infrastructures, such as traffic management systems, energy grids, and healthcare facilities, might reside in separate silos due to organizational boundaries or regulatory constraints. Cross-silo FL enables these disparate entities to collaboratively train a global model without directly sharing the raw data. This approach ensures that sensitive information remains confined within its respective silo and also encrypts the shared parameters among the collaborative organizations, while still allowing for the extraction of collective insights that can enhance the overall functionality and efficiency of the IoT system.

C. LOCAL LEARNING MODELS

The selection and deployment of ML models in FL clients are influenced by the specific training objectives and the nature of the data. For instance, object classification from images imposes the use of convolutional neural networks (CNNs) [29]. The federated stochastic gradient descent (Fed-SGD) is a predominant ML technique, often paired with neural networks (NN) for tasks such as image classification [29] and word prediction [78], [106]. Decision trees (DT) stand out as another widely adopted ML method, renowned for their efficiency in training. In the context of tree-based FL, models are crafted to train single or multiple DTs. Notably, research works [111], [112] have innovatively designed gradient-boosting decision trees (GBDTs) to cater to both horizontal and vertical data partitioning schemes. On the simpler end of the spectrum, linear models, including linear regression, logistic regression, and support vector machines (SVM) [113], [114], offer convenience in handling and learning, especially when compared to more intricate models like DTs and NNs. On the other hand, shallow neural networks, due to their fewer layers, are often favored for IoT devices with limited computational resources. RNNs and LSTMs, adept at capturing temporal patterns, find their niche in monitoring tasks like environmental conditions or machinery health. Autoencoders, adept at anomaly detection, are pivotal in machinery health monitoring scenarios. Fed-SGD, an optimization technique, is foundational to many FL applications and frameworks [29], [82], [115], [116]. While it is commonly associated with NNs, its versatility extends to models like SVM and linear regression. To bridge the gap between large-scale FL Systems (FLS) and cutting-edge ML models, it is imperative to harness the potential of ML architectures for enhanced FL training.

D. RESOURCES

Resources play a crucial role in determining the efficiency and effectiveness of the learning process particularly in the case of resource constraint devices. Broadly, these resources can be categorized into two main types: Computational Resources and Communication Resources.

1) *Computation Resources:* Computation resources refer to the processing power and memory available on the IoT devices participating in the FL process. Generally, IoT devices vary widely in their computational capabilities. While some devices, like smartphones or edge servers, might have significant processing power, others, like sensors or wearables, might have limited capabilities. The processing power determines how quickly a device can perform local model updates. Moreover, Memory is another crucial factor, especially when considering the size of the ML models and the data they process. Devices with limited memory might not be able to handle large models or might require model compression techniques to participate in FL. Similarly, for many IoT devices, especially those that are battery-powered, energy consumption is a significant concern. The computational tasks associated with FL can drain battery life, making it essential to optimize computations for energy efficiency.

2) *Communication Resources:* Communication resources are defined as the bandwidth and reliability of the communication channels used to exchange model updates and other information among the participating devices and the central server. Hence, the speed and capacity of the communication channel play a pivotal role in FL. High bandwidth allows for faster transmission of model updates, leading to quicker convergence of the global model. However, many IoT devices might be connected through low-bandwidth channels, making it essential to minimize the size of the transmitted data. In addition, the delay in communication can affect the synchronicity of the FL process. High latency can lead to delays in aggregating model updates, potentially affecting the learning process. Likewise, the stability and reliability of the communication channel are vital. Intermittent connections or high packet loss rates can disrupt the FL process. Techniques like local buffering or error correction can help mitigate these challenges. Furthermore, especially in environments where data

transfer costs are a concern, it is essential to minimize the amount of data exchanged. Techniques like model quantization or sparse updates can help reduce the data footprint.

E. PRESERVING PRIVACY MECHANISM

To maintain data privacy in FL-enabled IoT, various techniques has been applied. Differential Privacy (DP) adds noise to protect data, potentially reducing accuracy [117], [118]. Perturbation also secures privacy by altering data before analysis [119]. Homomorphic Encryption enables encrypted data processing, demanding significant computation [120]. Secure Multi-Party Computation (SMPC) facilitates confidential collaborative computing, though resource-intensive. Challenges persist across these techniques, including computational demands, security risks, data quality, and the absence of standardized FL protocols, impacting efficiency and interoperability in IoT [117], [118], [119], [120].

F. FL INCENTIVE MECHANISMS

Incentive mechanisms in FL for IoT are crucial to ensure the active participation of devices. Given the decentralized nature of FL and the diverse range of IoT devices with varying capabilities and interests, designing effective incentive mechanisms can be challenging. Several techniques, including game theory, matching theory, contracts, and regulations, have been proposed to address these challenges.

1) *Game Theory:* Game theory offers solutions like the Stackelberg game, where a leader (FL server) and followers (IoT clients) interact to find an equilibrium strategy such that both the server and devices have no incentive to deviate from their strategies. However, this interaction between the leader and followers can sometimes lead to non-converging strategies, making it hard to find a stable equilibrium. Auction-based method is another approach where IoT devices can bid for participation in FL based on their data quality, computational capabilities, etc. The FL server can then select a subset of devices based on their bids. Here devices bid for participation and might favor richer or more powerful devices, sidelining smaller participants which is a concern. Also, in the Cooperative Games approach, devices collaborate to achieve a common goal, e.g., improving the global model's accuracy. The reward is then distributed based on the contribution of each device.
2) *Matching Theory:* Matching theory is used to pair IoT devices with FL tasks based on preferences and capabilities, ensuring optimal learning. For instance, devices with specific types of data might be matched with FL tasks that require such data. Therefore, It ensures that the most suitable devices are chosen for specific FL tasks, optimizing the learning process. Nevertheless, this type of incentive approach can be computationally intensive, especially with a large number of devices and tasks. It might also lead to suboptimal matches if preferences aren't accurately represented.
3) *Contract Theory and Regulatory:* The interplay between contract theory and regulatory monitoring is frequently observed within FL. The contract theory is concerned with the development of formal agreements that establish the terms and conditions governing the relationship between the FL server and IoT devices. These agreements outline the specific obligations and benefits associated with their interaction. The purpose of these contracts is to address challenges such as moral hazard, wherein devices may not perform optimally, and adverse selection, which leads to a prevalence of low-quality devices in participation. In contrast, regulatory agencies play a crucial role in establishing criteria for FL participation, thereby securing equitable practices and conformity to established standards. Yet, both approaches encounter difficulties inside the decentralized FL environment. The decentralization of operations poses a substantial challenge in terms of monitoring and enforcing contractual agreements. In a similar vein, regulations, to uphold equitable competition, may occasionally exhibit excessive restrictiveness, impeding the progress of innovation. Additionally, the quick improvements in FL and IoT technologies can render these regulatory approaches outmoded, threatening their relevance and effectiveness.

G. FL NETWORK STRUCTURE

FL can be categorized based on network topology into three categories: Centralized, decentralized, and HFL [23].

1) *Centralized FL:* In this setup, all clients communicate directly with a central server. After computing model updates locally, they send them to this server. The server then aggregates these updates and dispatches the revised model back to the clients [26]. While this design is straightforward to implement, it might not be ideal for scenarios with a vast number of clients due to potential scalability issues. Moreover, having a single point of communication can introduce a single point of failure, making it less ideal for IoT applications that often experience intermittent connectivity.
2) *Decentralized FL:* On the other hand, Decentralized FL offers a more Distributed. approach. Here, clients communicate with each other in a peer-to-peer fashion, eliminating the need for a central server. They exchange model updates amongst themselves, and aggregation happens in a decentralized manner [121]. Each client aggregates updates from its neighbors. This structure is robust as there is no single point of failure. It is also potentially scalable to accommodate a large number of clients. However, the absence of a central coordinating entity might lead to slower convergence,

and the overall implementation can be more complex, especially in terms of peer discovery and management.

3) *Hierarchical FL:* This structure adopts a multi-tiered approach. In an IoT context, devices might first send their updates to an intermediary, like an edge server. Multiple edge servers then consolidate these updates and communicate with a central server for another round of aggregation [91]. This structure is particularly adept at handling a large number of clients and can manage intermittent connectivity at the device level by relying on edge servers. It also reduces the communication burden on the central server. However, the trade-off is the added complexity in setup and management, as coordination between different levels of hierarchy becomes essential.

H. FL OPTIMIZATION MECHANISMS

FL Optimization techniques have been specifically developed to enhance the performance of FL in terms of accuracy, convergence speed, and communication efficiency. It is also known as the aggregation algorithm as it is the logic that combines the local model updates from all the clients that participated in the training round [122]. The optimization process in FL presents significant challenges arising from non-IID data distributions, the presence of stragglers, communication overheads, and various other difficulties. In the context of the tasks they address, FL optimization algorithms can be categorized into single-task and multi-task.

1) *Single-Task:* In single-task FL, all devices collaborate to train a single global model on a shared single task. The primary goal is to aggregate local model updates in a way that the global model benefits from the collective data without accessing the raw data directly. Several widely used Single-Task FL optimization techniques are listed below: Federated Averaging (FedAvg) FedAvg was proposed by McMahan et al. [29] and is one of the most popular algorithms for FL. In this method, each client computes the average of its local model updates like gradient descent, and sends this average to the central server. The server aggregates these averages to update the global model. The updated global model is then sent back to the clients for the next round of training. Federated Stochastic Gradient Descent (FedSGD) In this approach, clients send their updates after processing each individual data point, rather than after a full local epoch. FedSGD typically converges more quickly than FedAvg. The reason is that FedSGD refreshes the global model with every step, whereas FedAvg does so after several steps. However, FedAvg tends to be more communication-efficient compared to FedSGD, as FedAvg permits devices to undergo multiple training steps before transmitting the updated weights to the server. In contrast, FedSGD mandates devices to forward their updated weights to the server following every step [29]. Federated Proximal Stochastic Gradient Descent (FedProx) This algorithm is an extended version of FedAvg and it introduces a proximal (regularization) term to the local loss function to handle non-IID data and stragglers, ensuring local models don't diverge too much from the global model [115]. The inclusion of the proximal term serves a dual purpose. Initially, it acts as a constraint, confining the modifications made by individual clients to mitigate the challenges stemming from statistical disparities in their datasets. Secondly, it facilitates the secure engagement of a variable number of clients in the collaborative learning process. Despite that FedProx was developed to address data and system differences it has two specific practical challenges associated with it: the difficulty in choosing appropriate weights for a term within the algorithm and the uncertainty about whether FedProx can guarantee an improved convergence rate for various ML applications [109]. These challenges suggest that while FedProx may be a promising technique, it still requires further research and development to overcome these practical issues for widespread adoption.

2) *Multi-Task FL:* In multi-task FL, devices collaborate to train multiple models simultaneously for different tasks [123]. The goal is to leverage shared representations and knowledge across tasks to enhance the performance of individual tasks. Federated Multi-Task Learning This approach trains multiple models for different tasks simultaneously, leveraging shared representations to enhance individual task performance [15]. In scenarios with a hierarchical structure, specialized algorithms optimize learning across different levels of hierarchy, addressing multiple tasks at different layers [124].

V. KEY CHALLENGES AND STATE-OF-THE-ART SOLUTIONS

Several challenges hinder the adoption of FL for IoT devices with limited resources. Table 8 summarizes these challenges. In the following sections, we discuss them in detail.

A. RESOURCE CONSTRAINTS IN IOT DEVICES

The implementation of FL in the network edge encounters considerable difficulties as a result of the limited resources of IoT devices. These devices, which are frequently characterized by their restricted processing power and low memory capacity, provide significant constraints, all of which have direct implications for FL operations. Firstly, IoT FL clients can exhibit severe limitations in memory size. Even in cases where a client possesses a larger memory capacity, it might not always have sufficient space to accommodate the data and model updates necessary for FL tasks. This discrepancy between memory size and storage requirements can pose significant challenges during the FL process. Secondly, FL clients in the IoT ecosystem often operate within predefined energy budgets. Unfortunately, these energy budgets may

TABLE 8. Summary of resource constraints in IoT devices and current solutions.

Reference	Challenge	Description	Proposed Solutions	Limitations
[125]	Memory Constraints	IoT devices often have limited memory, insufficient for FL tasks.	Distribution of data segments among clients with higher capacities; Sparse models (FedMask).	Sparse models may not be universally applicable.
[121]	Energy Budgets	Devices have predefined energy limits, not always suitable for FL.	Designing FL-specific hardware to reduce energy use.	High investment costs and not feasible for all devices.
[84]	Computational Capabilities	Limited computational power increases processing times.	Co-design of hardware and algorithms with IoT architecture considerations.	Complexity of co-design and unaddressed resource limitations.
[126]	System Performance	Additional communication overhead from constrained resources.	Hardware and algorithms to optimize efficiency and communication.	Optimization may not scale for all IoT deployments.
[127]	Hardware Utilization	Research on memory and energy use during FL is limited.	Exploring FL on devices like Raspberry Pi.	May not apply to all device types or larger systems.

not align with the rigorous system requirements demanded by the FL training process. This misalignment can hinder the ability of IoT devices to effectively participate in FL tasks. Furthermore, when IoT FL clients have limited computational capabilities, they are challenged with increased processing times.

In contrast, memory shortages can result in data overflow scenarios, which can overwhelm the device's resources and lead to poor performance. These resource-constrained situations introduce additional communication overhead into the FL process. Consequently, this extra communication overhead can have detrimental effects on the overall system performance, as it can impede the efficient exchange of data and model updates among IoT devices involved in FL.

Designing dedicated hardware tailored for FL is a promising approach to boost inference and training efficiency, significantly reducing energy consumption compared to generic processors, as highlighted by [78]. An example of this dedicated hardware for FL is FedMask [125] which represents an illustrative proposal in the domain of FL, focusing on both computational efficiency and communication optimization. FedMask introduces the concept of developing a heterogeneous and structured sparse binary mask on individual devices. This mask plays a pivotal role in generating a sparse model, subsequently leading to reductions in computation overhead, memory usage, and energy consumption. Nonetheless, it's worth noting that the FedMask approach remains agnostic to hardware specifics. To further alleviate the resource requirements associated with federated training, we anticipate that the collaborative synergy between hardware and algorithm design, emphasizing the incorporation of IoT hardware architecture considerations during the federated training process, represents a compelling avenue for future exploration. This co-design approach is a viable solution for achieving greater efficiency in FL while tailoring solutions to specific hardware constraints.

Considering the aforementioned limitations regarding memory capacity, authors in [121] introduced a method aimed at expediting the execution of the desired model in FL, given the constraints. Their approach involved the distribution of data segments among FL participants who exhibited larger memory capacities, enhanced energy budgets, greater bandwidth, and superior processing capabilities. Notably, their work did not extensively explore the complex elements of memory management and data handling for FL participants operating under the constraints of limited available memory. Memory constraints can be managed by storing data in small amounts. When memory is insufficient, data can be grouped together to prevent memory overflow. Study cases about challenges of using limited hardware resources in the implementation of FL, especially on devices like Raspberry Pi and other resource-limited clients, were investigated in [83], [84], [127], [128]. While they explored the possibility of using FL on devices with limited resources, they didn't delve into the best ways to use memory efficiently or measure energy consumption during the FL process.

B. COMMUNICATION OVERHEAD

In the FL environment, the central server distributes the global model to a potentially vast number of participating devices. During each communication cycle, these devices receive the model, refine it with their local data, and send the updated version back, imposing bidirectional data transfers [129]. However, challenges arise primarily due to limited bandwidth and device power, which can impede the speed and reliability of these communications [82], [115].

Unlike traditional setups relying on powerful GPUs and CPUs, computation in FL is now dependent on edge devices. These devices might face constraints in terms of computation, power resources, storage, and, as previously mentioned, limited link bandwidth [130]. This shift to edge devices introduces unique challenges, especially when considering the substantial differences in device capabilities. Another significant factor contributing to communication complexities is statistical heterogeneity. Specifically, when data is non-independent and non-identically (non-i.i.d.) distributed [115], it can lead to communication bottlenecks or increased communication costs. In FL, data resides locally on each device, collected based on individual usage patterns and local environments, further emphasizing the importance of understanding data distribution.

Transmitting model updates is more efficient than relaying large data volumes to a central hub, but there's an imperative to further optimize bandwidth usage [103]. The communication complexities are worsened by inefficiencies in server-client interactions. Clients, especially those with limited resources, face difficulties, whether it's connecting to the FL server or processing local tasks. The presence of extensive network data can also result in large model sizes, posing additional challenges. Various current methods place a strong emphasis on optimizing the efficiency and effectiveness of communication in FL [131], [137], [138], [139]. For instance, a study conducted by [82] explored two techniques: one that learns an update from a set space and another that updates and sends a compressed model to the central server.

The Communication-Mitigated FL (CMFL) approach [140] reduces unnecessary data transfers to the server by checking if local updates align with global ones, thus cutting down on communication overhead. The Federated Maximum and Mean Discrepancy (FedMMD) method [132] minimizes communication rounds by incorporating the Maximum Mean Discrepancy (MMD) constraint into its loss function. Also, authors of [81] introduced the Fed-Dropout method, an extension of the idea from [133], which creates a smaller version of the global model for local updates, reducing communication costs by exchanging these smaller models between the server and clients. The authors of [141] introduced FedDM, which generates synthetic data on the client side that mirrors the original data's distribution in the loss function. Sending this synthetic data, rather than model parameters, can cut down on communication overhead while conveying more information. Dai's group [142] employs a decentralized sparse training method, allowing each local model to use a unique sparse mask to pick its parameters and keep a consistent number of these active parameters during local training and direct communication. This means each local model only sends its active parameter indices once, and subsequent communications only involve these parameter values, not the entire model, significantly reducing overhead. Additionally, the layered decentralized FL structure can somewhat ease the communication burden on both server and client. Yet, it's not without its challenges, such as delays in multi-hop communication, uneven communication loads, and out-of-sync communication. Regular server-client communication in FL can speed up convergence but may be costly. An optimal method of balancing communication costs and FL resource usage is essential. While the authors of [143] studied these tradeoffs, they did not investigate the intricacies of clients' local model solutions.

Despite these research efforts, FL still faces challenges in communication efficiency. The presence of many edge nodes can elevate computational and storage demands. Variability in network bandwidth might cause delays or data losses when transmitting local models to the central server. Additionally, differing dataset sizes among clients can further delay model updates. Therefore, striking a balance between swift communication and model precision is essential.

C. FAIRNESS IN FL

Fairness refers to ensuring that the learning process and its outcomes are equitable across all participating devices or nodes. The data collected by these devices can be extremely different, leading to biases in the shared model. For instance, urban devices might have different data insights than rural ones. Additionally, the varied computational strengths of these devices mean that more powerful ones might overshadow the contributions of their less capable counterparts. Communication inconsistencies further complicate matters, as devices with unstable connections might struggle to contribute effectively. Moreover, not every device might be consistently available for every training round, leading to potential representation gaps in the global model. Lastly, security threats, such as biased data injections by malicious actors, can skew the model, affecting its fairness. Addressing these challenges is essential to ensure that FL truly benefits the entire IoT network and its users.

There have been research efforts to address fairness in FL [144], [145]. One notable novel approach is the work of [146], which is designed to optimize the centralized model's target distribution, formed as a blend of the distributions from participating clients. However, this method has primarily been tested on smaller scales. Also, Li et al. [134] introduced a method using an α fairness metric, termed q-Fair FL, aiming to ensure a balanced accuracy distribution. Their approach allows for the adjustment of resource distribution based on the desired fairness level. Another significant effort is a collaborative FL fair framework proposed in [135], which takes into account client reputation, pushing them towards different model outcomes. This method promotes fairness successfully without compromising the model's predictive capabilities. Additionally, research in [136] investigated an FL-based client selection process, aiming to reduce the time clients spend exchanging models. This approach ensures sustained and adaptable fairness, even with strict system limitations. However, they encountered challenges

TABLE 9. Communication and fairness challenges and current solutions.

Reference	Challenge	Under Category	Description	Proposed Solutions	Limitations
[82], [131]	Model Size	Communication.	Large network data increases model size, complicating FL.	Techniques for updates from a set space and compressed model transmission.	Compressed models may lose vital information, impacting the learning process.
[132]	Communication Rounds	Communication.	Extensive communication rounds are burdensome.	FedMMD minimizes rounds by integrating MMD constraint into the loss function.	MMD integration may not be suitable for all types of data distributions.
[133]	Communication Costs	Communication.	Costly model exchanges between server and clients.	Fed-Dropout uses smaller global models for local updates.	Smaller models might not capture the full complexity of the data. Management overhead
[134]	Accuracy Distribution Balance	Fairness	Ensuring balanced accuracy distribution across clients.	q-Fair FL method using an α fairness metric to adjust resource distribution for fairness.	Adjusting for fairness could potentially sacrifice client optimization.
[135]	Fair Model Outcomes	Fairness	Considering client reputation to push towards fair model outcomes.	Collaborative FL fair framework that accounts for client reputation.	It does not consider individual client needs.
[136]	Client Selection	Fairness	Reducing model exchange time to promote fairness.	May not apply to all device types or larger systems	Fairness factor affects the model's convergence rate and final accuracy.

in determining how the fairness factor might influence the model's convergence rate and final accuracy. After reviewing the approaches related to FL fairness, it becomes clear that clients operating within an FL-IoT framework could encounter resource limitations. Hence, developing a fairness system for distributing and allocating resources is vital to cut down on communication challenges, lessen computational overhead, and boost model accuracy. It's imperative to track client behaviors, estimate their resource capacities, and measure their impact on model progression to uphold fairness in FL resource management.

D. HETEROGENEITY IN FL

The heterogeneity within FL for IoT manifests in multiple forms, encompassing heterogeneity in data distributions, model architectures, network environments, and hardware devices among participating IoT clients. This heterogeneity underscores the varied nature of factors influencing IoT, ranging from the data characteristics to the technological infrastructure employed by the individual clients [147]. The most important heterogeneity types in FL-enabled IoT are data heterogeneity and system heterogeneity.

1) DATA HETEROGENEITY

Data heterogeneity is also known as statistical heterogeneity and it arises due to uneven data distribution among clients. This leads to a scenario where the data do not follow the same sampling pattern, resulting in what is known as non-IID data. This phenomenon poses unique challenges in FL. Current studies address the challenge of data heterogeneity using various approaches which are described as follows:

1) *Regularization:* To mitigate overfitting in FL models, a penalty is introduced in the loss function, which dynamically adjusts the complexity of the model. This strategy also helps to ensure convergence when learning from data that is diverse across different clients [148]. Building on the FedAvg [29] framework, FedProx [115] adds a term that narrows the gap between the client's local model and the global model, thereby enhancing stability and convergence. Moreover, to prevent the issue known as catastrophic forgetting, elastic weight consolidation [149] is adapted for FL scenarios. This technique judiciously penalizes the changes to parameters that are crucial based on the knowledge gained from previous

tasks. Further developments in regularization strategies have been proposed to tackle the challenges of client drift and model heterogeneity. For example, SCAFFOLD [150] corrects the update direction for both server and client models, while MOON [151] addresses the regularization between the global model and two sequential local models. To contend with noise and heterogeneity in client data, methods have been devised to apply weighted regularization based on the estimated noise level. This helps to improve performance for less advantaged clients by maintaining an average performance metric [152]. Additionally, proposals have been made for making sample embedding more consistent by aligning them with unbiased prototypes within the same domain, ensuring they remain free of irrelevant domain information [153]. An elastic aggregation approach [154] that adapts based on sensitivity analysis has also been introduced. This approach assesses the impact of parameter variations on the overall predictive function, acting as an implicit form of regularization.

2) *Clustering:* This is one of the most methods that addresses Data heterogeneity in FL. Basically, it involves a systematic grouping of clients based on various attribute similarities such as data distributions, the local models, and, the model updates of different clients [155], [156]. The work of Ghosh et al. [155] represents a groundbreaking research effort in CLF to overcome statistical heterogeneity in FL. It presents a method for clustering on the client side, wherein each client initially determines a standalone optimal solution locally. Subsequently, the clients transmit their models to the central server, which then groups the clients based on these locally optimized solutions. However, this method does not incorporate neural networks, based on the assumption that the algorithm's parameters are low-dimensional. Similarly, the work of [160] presents a hierarchical clustering method to tackle the challenges of statistical heterogeneity in FL, which impede convergence and escalate communication rounds. This method clusters clients according to the similarity of their updates to the global joint model. However, this study neglects to consider the diverse and significant aspects of model heterogeneity in its analysis. On the other hand, [157] offers a different approach with a one-shot clustering technique aimed at managing heterogeneity in FL. In this method, each device initially tackles a local clustering issue and calculates the average for that cluster. Subsequently, the central server collects these averages and combines them to establish the definitive cluster. Moreover, Additionally, Sattler et al. [161] introduce the concept of clustered FL, which tackles the non-IID problem by grouping clients into various clusters. However, the hierarchical clustering within the method is realized through several stages of bipartite separation, each requiring the federated SGD algorithm to reach convergence. As a result, its computational and communication efficiency may pose challenges in practical applications.

3) *Client Selection and Scheduling:* In FL, efficient client selection (CS) and scheduling play pivotal roles in enhancing communication efficiency and minimizing data heterogeneity [162]. Client selection (CS) involves carefully choosing devices with relevant data, reducing unnecessary data transfers, and mitigating communication overhead. Simultaneously, client scheduling optimizes when these selected clients participate, ensuring efficient utilization of available resources, which is paramount for minimizing communication costs. The synergy between these two concepts is instrumental in addressing the core challenge of data heterogeneity and communication efficiency in FL, contributing to faster convergence, improved model performance, and maintaining data privacy in decentralized settings [17]. Various methods have been developed to counteract the skewness caused by non-IID data in client selection. For instance, the work [158], driven by experience, is notable for its active identification and recruitment of the most suitable clients for participating in FL cycles. This approach treats device selection in FL as an RL challenge, aiming to train an agent to master an effective selection strategy. Furthermore, issues of class imbalance emerge when there's inconsistency in client data distribution. To tackle this, authors of [163] have formulated a method to determine client class distribution. This method relies on analyzing the gradient of client service updates, bypassing the need to access the original data. Additionally, they have crafted a client selection algorithm focused on minimizing class imbalance, thereby enhancing the convergence of the global model. Moreover, [164] introduced a client selection mechanism based on correlation, employing a Gaussian process to evaluate client loss variations. This method involves the choosing one client per iteration to minimize overall loss expectancy. Reference [165] argue that client selection for collaborative training should not only consider individual client characteristics but also the collective synergy among clients.

2) SYSTEM HETEROGENEITY

The variance in tasks and unique needs of clients leads to a situation where each client might prefer to craft their own local model [166], [167]. This individual approach, however, poses challenges in sharing knowledge across diverse participants. In such scenarios, commonly employed techniques like model aggregation or gradient operations become less effective due to the lack of uniformity among the models. Two primary approaches address model heterogeneity. A

TABLE 10. Heterogeneity challenges and current solutions.

Reference	Challenge	Description	Proposed Solutions	Limitations
[150], [151]	Client drift	The divergence of models on different devices due to varied data	Technique to correct the update direction for both server and client models.	Sub-optimal performance in scenarios with rapidly changing data distributions.
[152]	Noisy data	irrelevant or random information that can distort the data	Weighted regularization based on the estimated noise level.	Inaccurate noise estimation leads to improper regularization, negatively impacting model performance.
[155], [156]	Data heterogeneity	Variance in data distribution across clients, causing challenges in model training due to non-IID data	A method that clusters clients according to the similarity of their updates to the global joint model	Computationally intensive with a large number of clients
[157]	Data heterogeneity	Variance in data distribution across clients, causing challenges in model training	The one-shot clustering technique aggregates local cluster averages from devices to form a final cluster in FL.	Reduced accuracy in highly diverse datasets, as it relies on a single aggregation step to form the final cluster.
[158]	Skewness	Uneven distribution of data across clients, resulting from non-IID data characteristic.	DRL method that trains an agent to master an effective client selection strategy.	Fairness issue
[136], [159]	System Heterogeneity	Variation in computational power, memory capacity, and network connectivity among the devices participating in the FL process	KD method simplifies complex model insights for efficient learning on diverse devices	Reliance on public datasets and security risks from protocol incompatibility.

brief overview of both approaches is given in the subsequent sections.

1) *Knowledge Distillation (KD):* KD is a key approach in addressing model diversity within FL, especially in cross-device environments such as IoT and mobile devices [159]. For instance, FedDF [168] harnesses KD by extracting insights from various classifiers, each trained on private data from a network of client devices. It uses these classifiers' responses to an unlabeled, shared dataset to guide the training of a centralized 'student' model via KD. Similarly, DS-FL [169] adopts a distillation-based, semi-supervised approach in FL. It improves performance by applying pseudo-labeling to public data at the server level. In contrast, FedGKT [170] introduces a group knowledge transfer method, enabling knowledge to flow from client devices to a larger server model, even in the absence of public data. Additionally, Fed-ET [171] innovates with a weighted consensus distillation approach, coupled with diversity regularization. This method facilitates training a comprehensive server model using smaller client models. However, KD-based methods are not without drawbacks. A significant limitation is their reliance on public datasets for attaining competitive accuracy levels. The effectiveness of these methods is closely tied to the volume and domain relevance of the public data in relation to the client data [168], [171], [172]. Moreover, since KD often uses client model weights, either partially or wholly, as a teaching mechanism for server knowledge transfer, this approach is often incompatible with secure aggregation protocols. Such incompatibility increases susceptibility to security risks like backdoor attacks [173].
2) *Partial Training (PT):* PT is becoming increasingly used as a practical approach for handling heterogeneity in models in FL. PT-based methods generally fall into two categories based on the technique used for extracting sub-models from the overarching server model: random and static sub-model extraction. Taking a cue from the dropout method prevalent in centralized training [133]. The work of [81] introduces a random

extraction of sub-models from the global model. While this method integrates seamlessly into current FL systems, its effectiveness diminishes with high data heterogeneity and a smaller client cohort, as noted in [174]. This is primarily due to the random nature of sub-model selection. On the other hand, methods like HeteroFL [175] and FjORD [176] employ a static extraction approach. Here, sub-models are consistently drawn from a specific portion of the global server model. However, this static strategy comes with notable limitations. Firstly, it binds the global server model to the size of the largest client model. This dependence curbs the size and capabilities of the global model, often limiting its ability to manage larger models due to resource constraints on the client side. Secondly, and more critically, static extraction means that different sub-models are confined to being trained on clients with corresponding resource capacities. This results in parts of the global server model being unable to train on data from less powerful client devices. Such a scenario leads to varying training data distributions across different parts of the global model, which can significantly hinder the performance of the global model in scenarios with high data heterogeneity.

E. SECURITY AND PRIVACY IN FL

Privacy is now commonly acknowledged to be of utmost importance within the framework of FL [109]. The main reason for this is that protecting the local data of clients from potential intrusions is inherent to the fundamental principles of FL. During FL communication, clients avoid sharing private data, respecting privacy principles [103]. Yet, the nature of FL doesn't fully prevent data breaches due to hidden vulnerabilities and heterogeneity that can exacerbate privacy issues. For instance, adversaries might deduce sensitive information during FL participation by manipulating data or analyzing gradients, potentially leading to unintended data exposure to servers or other clients. To counter privacy threats, differential privacy (DP) techniques, such as adding noise to sensitive attributes and Gaussian noise to local model gradients, are employed to safeguard personal information and ensure data integrity while maintaining model convergence [187], [188], [189], [190]. Notably, authors of [189] integrated both Homomorphic Encryption (HE) and DP in their approach. A notable challenge arises when fewer clients engage in the DP mechanism; it can lead to significant distortions in the combined model. This presents a dilemma between maintaining privacy and ensuring model precision. To address the decline in accuracy, authors in [191] introduced a novel normalization method in their neural networks, deviating from the conventional batch normalization seen in studies like [189] and [187]. Apart from Multi-Party Computation (MPC) and DP, a study by [192] adopted a distinct method for safeguarding datasets. Rather than clients distributing obscured gradients, the FL Framework hinges on requesters disseminating concealed datasets during the model's verification phase. This strategy obstructs other participants from replicating the models during their assessment and testing phases. The techniques of HE and 2-party computation (2PC) are employed in this context. Other researchers, such as [177], [178], [179], [193] have also emphasized the concealment of gradients. However, they have not detailed the specific methods they used to preserve privacy. Additionally, a method that combating data poisoning attacks involves data cleaning methods is proposed in [194]. It uses an influence function to assess and mitigate the adverse effects of erroneous data on model performance, thereby enhancing the efficacy of FL systems.

To counter model poisoning attacks, defenses focus on robust aggregation and anomaly detection. Robust aggregation includes techniques like Byzantine-robust aggregation rules [180], [181], [182], [195]. FLTrust [182], for instance, involves a server using a small clean dataset to participate in updates and discard gradients that significantly deviate from its own, enhancing accuracy by normalizing gradient magnitude. Anomaly detection is represented by divide-and-conquer (DnC) [183], which employs spectral methods for identifying malicious gradients and uses random sampling to minimize memory and computational demands. These approaches, noted for their theoretical rigor, serve as key defense mechanisms against attacks. Moreover, To protect against model poisoning and free-rider attacks from within, auditing mechanisms are employed [103]. For example, spectral anomaly auditing with VAE [184] and a high-dimensional detection method [185], have been developed to identify malicious behavior. However, these techniques struggle to accurately pinpoint all malicious clients, especially in scenarios with multiple attackers, and risk high audit errors that could exclude honest clients from training. To enhance FL training reliability, [186] introduced a parameter audit method requiring clients to submit all training-relevant data for review, effectively identifying attackers with greater accuracy but significantly increasing system overhead.

VI. EMERGING TRENDS IN FL

As FL continues to advance, several novel trends are emerging that expand the capabilities and applications of decentralized learning. These trends address critical challenges and introduce innovative solutions, positioning FL as a key technology in the future of artificial intelligence and communication networks. Below, we explore some of the most cutting-edge trends in detail.

A. FL FOR FOUNDATIONAL MODELS

Foundational Models (FMs) are pre-trained on extensive, generic datasets to develop a broad set of capabilities, which can then be fine-tuned for specific tasks [196]. This lifecycle begins with pre-training, where the model learns from large-scale data to establish its foundational knowledge [197]. The model is then adapted for specific applications, such as domain-specific question answering [198], allowing it to

TABLE 11. Security challenges and current solutions.

Reference	Challenge	Description	Proposed Solutions	Limitations
[177]–[179]	Gradient concealment	Avoiding data exposure by concealing gradients	Gradient concealment emphasized	Privacy methods not detailed
[180]–[182]	Model poisoning	Attacks manipulating updates	Byzantine-robust aggregation, FLTrust	Difficulty pinpointing malicious clients
[183]	Anomaly detection	Identifying malicious gradients	DnC with spectral methods	High audit errors risk
[184], [185]	Auditing mechanisms	Detecting internal malicious behavior	Spectral auditing, high-dimensional method	May miss multiple attackers
[186]	Enhancing training reliability	Free-rider attacks and poisoning	Parameter audit method	Increased system overhead

be effectively applied across various fields. For instance, the rise of FMs such as Generative Pre-trained Transformer (GPT) [199] and Bidirectional Encoder Representations from Transformers (BERT) [200] has sparked interest in applying FL to train and fine-tune these models across decentralized data sources. These foundational models are built upon deep neural network (DNN) architectures, particularly on Transformer models [201], which have significantly transformed the natural language processing (NLP) field since their inception. A central feature of Transformer models, and by extension many FMs, is the self-attention mechanism. Moreover, the Transformer model features an encoder-decoder structure, which accepts tokenized data as input. This architecture consists of multiple layers, each known as a Transformer block. These blocks include two principal sub-layers: a multi-head self-attention (MHSA) mechanism and a position-wise, fully-connected feed-forward network (FFN), complemented by normalization and shortcut connections. Following the development of various foundational models, the Vision Transformer (ViT) [202] emerged as a groundbreaking FM approach for Computer Vision (CV). It extends Transformer encoders to image processing and has been widely adopted in tasks like recognition [203], detection [204], segmentation [205] performing well in both supervised [203] and self-supervised learning [206], [207], [208]. Recent research has also enhanced the theoretical understanding of ViT, particularly regarding representation robustness [209]and latent propagation [210], [211]. A significant advantage of applying FL to FMs is its ability to enhance data availability. FL facilitates the integration of personal and task-specific data, enabling FMs to be tailored for individual applications. For example, Google has utilized FL to train next-word prediction models using mobile keyboard input data, thereby improving the user experience [212]. Additionally, FedBERT [213] leverages both FL and split learning to pre-train the BERT model in a federated manner, ensuring data privacy while achieving strong performance without sharing raw data. Similarly, FedIT [214] uses FL to tune large language models (LLMs), leveraging user-generated data while addressing privacy and transmission costs. This study shows FedIT's effectiveness in enhancing LLM performance through qualitative analysis, including GPT-4 auto-evaluation. Furthermore, FL faces challenges like handling large, heterogeneous data, resource-heavy training, and high communication overhead. Authors of [215] address handling large, heterogeneous data, resource-heavy training, and high communication overhead in FL by exploring the learning stages of LLMs in wireless networks. They introduced personalized fine-tuning methods, personalized federated instruction tuning (PFIT), and personalized federated task tuning (PFTT). PFIT uses RL for personalized tuning, while PFTT leverages global adapters and local LoRAs for efficient, personalized tuning without aggregation.

While vision transformers have established themselves as the standard architecture for centralized computer vision tasks, their application in FL is still in its early stages. For instance, [216] assessed the performance of vision transformers in global FL settings, highlighting their robustness of data heterogeneity. Concurrently, FedPerFix, [217] and FedTP [218] explored the personalization of vision transformers within personalized FL frameworks. Additionally, research focusing on other aspects of vision transformers, such as pre-trained models, has also started to attract attention [219], [220], [221]. The efficiency challenges in FL stem from the resource-intensive nature of training FMs on systems with limited and heterogeneous resources, compounded by communication bottlenecks and a large number of model parameters. To address this challenge, several techniques have been proposed, including *Fine-Tuning (FT* which includes Instruction Tuning (IT), Task Tuning (TT) [215], [222], [223], [224], *Model Compression* [127], [225] and *Zeroth-order optimization* [212], [226].

B. FL FOR GREEN AND SUSTAINABLE 6G IOT

Green computing emphasizes designing and using computing resources to minimize energy consumption and reduce the carbon footprint and other things that impact the environment [227]. While Sustainable IoT, ensures the development and deployment of energy-efficient and environmentally friendly IoT systems, supporting long-term viability without compromising performance [228]. However, both usually

are used interchangeably [229]. Hence, the need for green sustainable 5G and 6G IoT solutions has become increasingly pressing as the world grapples with climate change and resource depletion challenges [230], [231]. The commercialization of 5G in the early 2020s has been highly successful, driven by innovations such as user-centric architecture [232], massive MIMO [233], and mmWave communication [234], which has significantly improved throughput, latency, and energy efficiency compared to 4G. While 5G has enabled advanced IoT services and enhanced user experiences across various domains, the rapid growth in service demands and the number of connected IoT devices are expected to outpace 5G capabilities. Consequently, 6G networks are anticipated to emerge by 2030, offering a seamless integration of space-air-ground and undersea networks, with data rates up to 10 Gbps, near-perfect reliability, and ultra-low latency for the Internet of Everything applications [235]. To achieve this, 6G will leverage breakthrough technologies such as THz and visible light communications, Reconfigurable Intelligent Surfaces, and fully integrated Endogenous AI, which will be essential at every stage of 6G development, from architecture to communication optimization [236]. Additionally, the distributed and heterogeneous nature of 6G networks will necessitate deploying AI at the network edge to enable edge intelligence. However, the limited capabilities of edge devices as a key enabler of green and sustainable IoT systems within the 6G ecosystem present challenges for training AI at the edge, which the FL aims to address [21], [237].

It has been observed that autonomous vehicles represent one of the most promising domains for 6G technology. For instance, the work in [238] introduces a hierarchical mobile-edge intelligence framework for automated pothole detection and intelligent road systems, leveraging FL to maintain data privacy and optimize model performance. This framework integrates diverse sensor inputs with a novel DL approach, combining image and sensory data to enhance detection accuracy and efficiency. Additionally, it incorporates 6G and AirComp technologies for seamless communication across road networks, enabling real-time alerts and resource allocation for road maintenance, thereby improving safety and operational efficiency. Similarly, Khan et al. [239] propose an FL framework for autonomous driving cars to overcome challenges in FL, such as robustness, server security, communication constraints, and privacy risks. The framework incorporates a mixed-integer non-linear programming (MINLP) optimization to reduce model accuracy loss due to packet errors and transmission delays. To solve the NP-hard problem, the authors introduce a Block Successive Upper-bound Minimization (BSUM) approach, demonstrating its effectiveness through comparisons with baseline schemes.

C. DEEP REINFORCEMENT LEARNING

RL utilizes an intelligent agent to make decisions aimed at optimizing rewards to achieve a specific objective [244]. It shares similarities with dynamic programming but differs from supervised and unsupervised learning by utilizing distinct policies during the learning process. The core aim of RL is to learn a policy that enables the agent to make optimal decisions within a given environment. The agent follows a reward-based learning approach, generating data through iterative interactions with the environment and continuously striving to maximize positive rewards. The environment is often modeled as a Markov decision process, where rewards and state transition probabilities depend on the agent's observations and selected actions. Ultimately, the goal of RL is to determine actions that maximize the accumulation of future rewards. The agent's performance improves progressively through repeated learning cycles [241]. RL traditionally requires manual engineering of the state space, which can be a complex and time-consuming process. DRL integrates DL into RL to help the agent make optimal decisions from unstructured inputs, effectively eliminating the need for manual state space design. The primary advantage of integrating DL into RL lies in the scalability of deep neural networks (DNNs) when handling high-dimensional spaces. For instance, in value function approximation, DNNs leverage their ability to represent complex data distributions, using end-to-end gradient-based optimization to capture these relationships effectively [240].

In FL environments, FedAvg is a key technique that iteratively selects a random subset of participants to perform local training. After distributing the global model, each selected participant trains the model locally, and the results are aggregated to update the global model. This process enables efficient model training without requiring data centralization. To optimize this framework, various factors such as cross-device settings, differential privacy, and communication types must be fine-tuned [242]. DRL can significantly enhance this process by converging FL initialization functions with DRL state features, such as the capacities of edge devices. The actions of DRL agents can be aligned with participant selection in FL rounds, while rewards are tied to the efficiency of federated optimization, opening new possibilities for real-world applications [245], [246].

DRL is particularly useful for managing dynamic resources in IoT environments. In FL-based IoT networks, DRL helps allocate resources efficiently by analyzing resource states and task sizes [247]. This ensures optimal decisions regarding resource allocation and task offloading, even in asynchronous or high-latency environments [243]. In such setups, the DRL agent continuously monitors system states, including transmission power and device resources, to make decisions about which devices should participate in each training round. The agent's actions are driven by the objective of minimizing energy consumption and maximizing the accuracy of the global model [248]. Moreover, DRL-based scheduling is effective for managing large-scale IoT environments by improving convergence speed and addressing straggler issues. By clustering devices based on their performance metrics, DRL can optimize local and global training updates. This is particularly useful in scenarios

TABLE 12. Summary of emerging trends in FL.

Reference	Trend	Aim	Advantages	Limitations
[196]	FMs	Apply FL to train and fine-tune foundational models across decentralized data sources.	Improves model adaptability and personalization.	Resource-intensive training with large model parameters.
[197]	FMs	Leverage pre-trained models for various tasks in decentralized systems.	Expands data availability and task-specific model tuning.	Communication bottlenecks during FL training.
[199]	FMs	Improve language models like GPT by incorporating decentralized learning.	Enhances personalization of models for specific applications.	Requires significant computational resources for training.
[200]	FMs	Fine-tune large models like BERT using FL to address decentralized language data.	Increased accuracy for specific tasks across multiple domains.	High communication costs and model parameter sizes.
[201]	FMs	Adapt transformer models for decentralized training and improve NLP applications in FL settings.	Efficient handling of unstructured input data with transformer architectures.	Limited application in FL environments due to computational requirements.
[227]	GS 6G	Develop energy-efficient IoT systems using FL to reduce environmental impact.	Minimizes energy consumption and promotes sustainable practices in IoT.	Limited computational power of edge devices.
[228]	GS 6G	Enable sustainable IoT deployments with FL, focusing on energy and resource-efficient communication systems.	Supports long-term viability of IoT systems while maintaining performance.	Edge intelligence faces resource limitations and communication constraints.
[235]	GS 6G	Leverage 6G networks for advanced IoT services, ensuring real-time communication with FL for enhanced decision-making.	High data throughput and ultra-low latency enable robust IoT applications.	Complex integration of space-air-ground networks with AI-driven edge intelligence.
[237]	GS 6G	Address the challenges of edge AI in 6G networks, particularly in resource-limited environments, using FL.	Enhances privacy and optimizes resource usage in edge networks.	Limited edge device capabilities hinder the training of complex models.
[240]	DRL	Use DRL to optimize decision-making in FL environments, enhancing resource allocation and participant selection.	Efficient handling of high-dimensional data for improved decision-making.	Requires careful design to avoid scalability and resource bottlenecks.
[241]	DRL	Integrate DRL into FL to improve optimization in dynamic IoT environments, particularly for resource management.	Provides adaptive resource allocation and task scheduling in FL settings.	High computational demands and complexity in large-scale environments.
[242]	DRL	Optimize FL processes by using DRL to manage communication and task offloading in IoT systems.	Increases system efficiency and reduces communication costs.	Potential straggler issues and delays in large-scale implementations.
[243]	DRL	Apply DRL in FL environments for IoT to optimize power management and task scheduling.	Efficient energy use and improved model accuracy in IoT systems.	Managing high-latency environments and ensuring low energy consumption simultaneously.

such as healthcare, where efficient resource allocation and trust management are critical [249]. DRL-based optimization techniques have also been developed for industrial IoT (IIoT) applications, focusing on task offloading, bandwidth allocation, and transmission power management. These approaches aim to reduce communication costs, stabilize energy consumption, and improve overall FL performance, ensuring robust and scalable IoT systems [250].

VII. FL FRAMEWORKS AND BENCHMARKS

Recognizing the potential and growing interest in FL, various organizations, and researchers have dedicated efforts to develop open-source frameworks to implement FL over IoT Networks [254]. These frameworks and libraries streamline communication between decentralized nodes, offer aggregation methods, and ensure robust security, providing diverse FL scenarios and needs. The most popular FL frameworks and libraries are listed as follows:

A. PYSYFT

PySyft [251] is an advanced framework built on PyTorch that emphasizes encrypted and privacy-preserving models. It incorporates techniques like SMPC and DP to operate safely in untrusted environments. While Pysyft has these added functionalities, it maintains the familiar PyTorch tensor operations, seamlessly integrating them. For FL simulations, PySyft uses "Virtual Workers" to represent

TABLE 13. Comparison of FL frameworks.

Framework	Developed by	OS	Key Features	Underlying Technologies	Suitability	Benchmarking
PySyft [251]	OpenMined	Linux, Win, Mac, iOS	Privacy-preserving DL, SMPC and DP, Virtual Workers, PointerTensor	PyTorch	Suitable for environments requiring encrypted and privacy-focused DL	✓
TFF [252]	Google	Linux, Win Mac	Federated Core API, FL API, Integration with TensorFlow models	TensorFlow	Ideal for TensorFlow users, flexible for both high-level and granular experiments	✓
LEAF [253]	CMU University	Linux, Win, Mac	Benchmarking datasets like FEMNIST and Sentiment140, Data partitioning based on individual participants	TensorFlow	Best for benchmarking FL algorithms	✓
FedML [254]	FedML	Linux, Win, Mac	FedML-core and API, Training and communication modules, Topology Manager	PyTorch	Suitable for developing new FL algorithms	✓
FATE [255]	WeBank	Linux. WSL	Focus on security and privacy, Federated execution of ML models	Pytorch	Ideal for scenarios where data privacy and security are paramount	×
Flower [256]	University of Oxford	Linux, Win, Mac	Framework-agnostic, scalable, supports diverse clients, privacy-focused, efficient communication	Compatible with TensorFlow, PyTorch, etc.	Suitable for both research and production, adaptable to diverse environments	×

participants. Data is divided among these workers, with a special "PointerTensor" indicating the owner of the data owner and location. This structure allows for efficient model updates and global aggregation.

B. TENSORFLOW FEDERATED (TFF)

TFF [252] is another framework developed by Google. Its primary purpose is to facilitate experiments with ML models on distributed datasets and other related computations. TFF offers a diverse range of interfaces, primarily segmented into two categories: the Federated Core API and the FL API. The FL API stands out with its high-level interface features, which seamlessly integrate with pre-existing TensorFlow models. This makes it an excellent choice for researchers and developers who already have TensorFlow models and wish to venture into FL without having to build everything from scratch. On the other hand, the Federated Core API delves deeper, offering a more granular set of interfaces. This is tailored for those looking to create new FL algorithms from the ground up. It provides the necessary tools and capabilities to design these algorithms using TensorFlow and manage distributed communication processes effectively. In essence, the Federated Core API provides more flexibility and control for those aiming to explore new horizons in FL using TensorFlow.

C. LEAF

LEAF [253] is a framework that contains many datasets specifically designed to act as benchmarks in FL scenarios. Among the noteworthy datasets within LEAF is the Federated Extended MNIST (FEMNIST). This dataset is a specialized version of the renowned MNIST dataset but with a twist: data partitioning is done based on the individual writer of each character. Another standout dataset is Sentiment140, which is segmented according to individual users. These partitions are highly significant. The writer in FEMNIST or the user in Sentiment140 is conceptualized as a participant in the FL process. The data corresponding to each participant is visualized as the local dataset residing on their respective devices. When researchers and developers implement their novel FL algorithms on datasets like these, it sets the stage for consistent and trustworthy comparisons across different studies, ensuring a standardized benchmarking process.

D. FEDML

FedML [254] is a new open research platform designed to facilitate the progress of FL. It comprises two principal components: FedML-core and FedML API. Within FedML-core, the model training and distributed communication processes are distinctly separated into the training module and the distributed communication module. The latter handles the intricate communication between different clients, while the training module is crafted using PyTorch. A client-centric programming interface allows for effortless integration of new algorithms in FedML. One of its standout features is the TopologyManager, which offers support for an array of network topologies, enabling the implementation of diverse FL algorithms.

E. FATE

Federated AI Technology Enabler (FATE) is an open-source platform that was developed by WeBank [255]. The primary objective of this cutting-edge platform is to facilitate the federated execution of ML models, with a particular focus on implementing robust security measures and privacy-preserving techniques. This approach ensures the preservation of data privacy and security. The architectural design not only facilitates the implementation of decentralized ML but also provides businesses with the ability to utilize collective data insights while upholding stringent data protection protocols.

F. FLOWER

Flower [256] is an open-source FL framework developed by various institutions and notable for its compatibility with various machine-learning tools, including popular frameworks like TensorFlow and PyTorch. Its design focuses on scalability and flexibility, allowing it to adapt to different client environments effectively. This makes Flower an ideal choice for both research purposes and practical, real-world applications. A key aspect of Flower is its strong emphasis on privacy and efficient communication, essential in FL scenarios where data security and network efficiency are crucial. It supports the integration of a wide range of client devices, even with differing data and hardware capabilities, while ensuring that sensitive information is processed locally and not centralized. These attributes render Flower a highly versatile and user-friendly framework, suitable for a broad spectrum of FL tasks.

VIII. OPEN ISSUES AND FUTURE RESEARCH DIRECTIONS

In earlier discussions, we explored the dynamic landscape and significant potential of FL in IoT systems. Despite this, the application of FL in IoT is still emerging, and there is a substantial need for research to develop an effective framework for its implementation in IoT environments. This section emphasizes the existing research gaps and identifies key areas for future investigations, building on the challenges previously mentioned.

A. MULTIMODAL FEDERATED LEARNING

As edge computing progresses, an extensive range of sensors and devices are increasingly capturing data from diverse modalities, such as tactile feedback from industrial equipment, thermal imaging for environmental assessment, and acoustic signals from urban environments. Consider autonomous vehicles in the automotive sector, which amalgamate data from lidar, radar, and cameras to refine navigation and safety systems. This variation in data sources creates environments where certain clients handle multiple data modalities (multimodal clients), whereas others manage only a single type (unimodal clients). The varied capabilities and resources of these devices highlight the imperative need for further investigation into Multimodal Federated Learning (MMFL) as a future direction for FL-enabled IoT.

B. ROBUSTNESS AND FAULT TOLERANCE FL

The integration of FL in the expanding domain of IoT presents significant problems in terms of robustness and ability to withstand faults. The deployment of IoT devices for various applications is on the rise due to their diverse nature, limited processing capabilities, and varied connectivity. Nevertheless, these attributes also render them prone to data discrepancies, security vulnerabilities, and operational failures. Although the decentralized structure of FL is advantageous for privacy and scalability, worsens these problems by dispersing the learning process among possibly untrustworthy nodes. The primary challenge is to guarantee the continued effectiveness and dependability of the FL system within this diverse and frequently volatile setting. To ensure continuous and reliable operation in diverse IoT applications, future research should concentrate on developing robust and fault-tolerant algorithms for FL in IoT environments. The primary focus should be on creating resilient operational frameworks. This includes Implementing robust data cleaning and preprocessing to handle noisy and incomplete data. Using adaptive anomaly detection techniques to identify and mitigate the impact of outliers or corrupted data. and Incorporating adversarial examples in training to improve the model's resilience against attacks. Moreover, researchers must propose techniques that regularly save the state of the model so that it can resume from the last checkpoint in case of interruptions and Design FL systems that degrade their functionality progressively under fault conditions, rather than failing completely.

C. ADAPTIVE RESOURCE ALLOCATION AND SCHEDULING

This concept refers to dynamically distributing and managing computational resources among various IoT devices. It includes optimizing battery usage, computational power, and network bandwidth. Despite that many works have been presented to address this issue but still it remains a complex challenge to the diversity of devices, energy constraints, and network limitations. Research should directed into developing FL algorithms that can efficiently manage data transmission, reducing the network load and accommodating varying bandwidth and latency conditions. Also, future works must develop scheduling algorithms that take into account the context of the IoT environment, such as the location of devices, mobility patterns, network conditions, and the type of tasks being performed. Another approach is to Prioritize tasks based on the energy availability of devices. For battery-operated IoT devices, scheduling less critical tasks during times of lower energy availability can conserve battery life.

D. ADAPTIVE DEFENSE AGAINST ADVERSARIAL ATTACKS

Research on cyberattacks impacting FL, particularly decentralized (DFL) and hierarchical FL (HFL), is scarce, with adversarial attacks receiving limited focus. Understanding the impact of these attacks on DFL and HFL, compared

to more traditional centralized FL (CFL), is crucial for identifying vulnerabilities within these architectures. This insight is essential for developing enhanced security measures. To address this issue, developing adaptive defense mechanisms that can dynamically adjust to the evolving nature of adversarial attacks is key. Such defenses include real-time detection algorithms, and automatic adaptation to attack patterns to safeguard data and model integrity.

E. UNSUPERVISED LEARNING IN FL

IoT devices generate vast amounts of unlabeled data, which is time-consuming and costly. The literature largely overlooks the potential of unsupervised learning in FL, where federation participants may not have locally labeled data. Most studies focus on supervised ML models such as MLP, CNN, and SVM, neglecting unsupervised approaches. Exploring unsupervised learning techniques specifically for FL can efficiently leverage unlabeled IoT data. Adopting semi-supervised and self-supervised learning models can reduce dependence on labeled datasets, while AI-driven tools for automatic data labeling can minimize manual effort and costs. Additionally, cross-domain data sharing and synthetic data generation can enhance model training without labeled data.

F. TOWARDS QUANTUM-ENHANCED FL

Quantum computing leverages quantum mechanics principles to outperform traditional computing systems in solving certain ML tasks with better speed and convergence [257]. Incorporating such quantum capabilities into FL is set to revolutionize the domain from diverse aspects [258], [259]. Quantum FL research for IoT may focus on deploying quantum devices at each node, enabling Quantum ML tasks for enhanced performance. Initially, quantum capabilities may be centralized on servers due to the immobility of advanced quantum computers such as superconducting qubits and ion-trap systems. This setup aims to accelerate training and aggregation processes. As quantum technology becomes more accessible, a transition towards fully quantum FL systems is anticipated, where all nodes possess quantum computing abilities.

IX. CONCLUSION

This paper investigated the state of the art of FL-enabled IoT systems, offering a detailed exploration of the synergy between FL and IoT. We began by providing a brief overview of the FL concept by focusing on its components and how they work together. Then, we presented the key features and advantages that FL brings to IoT systems. This is followed by a series of in-depth examples illustrating the application of FL in IoT systems, highlighting cutting-edge applications in healthcare, smart cities, and autonomous vehicles, and analyzing current trends and advancements. Furthermore, the paper presented a comparative analysis of different IoT-FL architectures, highlighting their unique characteristics and functionalities. A comprehensive taxonomy of FL-enabled IoT systems was then outlined, providing a structured insight into this field. The paper also addresses various challenges associated with integrating FL into IoT systems, offering insightful current solutions to these issues while highlighting the key limitations of these solutions. The existing FL frameworks were compared and contrasted. Finally, the review identifies open issues and future research directions, underscoring the dynamic and evolving nature of FL for IoT systems. The derived observations show how FL can significantly revolutionize the IoT landscape emphasizing the paramount importance of continuous research and interdisciplinary cooperation.

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REFERENCES

[1] J. Lin, W. Yu, N. Zhang, X. Yang, H. Zhang, and W. Zhao, "A survey on Internet of Things: Architecture, enabling technologies, security and privacy, and applications," *IEEE Internet Things J.*, vol. 4, no. 5, pp. 1125–1142, Oct. 2017.

[2] P. K. Sadhu, V. P. Yanambaka, and A. Abdelgawad, "Internet of Things: Security and solutions survey," *Sensors*, vol. 22, no. 19, p. 7433, 2022.

[3] I. Stellios, P. Kotzanikolaou, M. Psarakis, C. Alcaraz, and J. Lopez, "A survey of IoT-enabled cyberattacks: Assessing attack paths to critical infrastructures and services," *IEEE Commun. Surveys Tuts.*, vol. 20, no. 4, pp. 3453–3495, 4th Quart., 2018.

[4] M. Adam, M. Hammoudeh, R. Alrawashdeh, and B. Alsulaimy, "A survey on security, privacy, trust, and architectural challenges in IoT systems," *IEEE Access*, vol. 12, pp. 57128–57149, 2024.

[5] Y. Sun, M. Peng, Y. Zhou, Y. Huang, and S. Mao, "Application of machine learning in wireless networks: Key techniques and open issues," *IEEE Commun. Surveys Tuts.*, vol. 21, no. 4, pp. 3072–3108, 4th Quart., 2019.

[6] "General data protection regulation (GDPR)," Intersoft Consulting, 2018. [Online]. Available: https://gdpr-info.eu/

[7] O. Gómez-Carmona, D. Casado-Mansilla, F. A. Kraemer, D. López-de Ipiña, and J. García-Zubia, "Exploring the computational cost of machine learning at the edge for human-centric Internet of Things," *Future Gener. Comput. Syst.*, vol. 112, pp. 670–683, Nov. 2020.

[8] S. Singh, S. Rathore, O. Alfarraj, A. Tolba, and B. Yoon, "A framework for privacy-preservation of IoT healthcare data using federated learning and blockchain technology," *Future Gener. Comput. Syst.*, vol. 129, pp. 380–388, Apr. 2022.

[9] H. Elayan, M. Aloqaily, and M. Guizani, "Sustainability of healthcare data analysis IoT-based systems using deep federated learning," *IEEE Internet Things J.*, vol. 9, no. 10, pp. 7338–7346, May 2022.

[10] H. Elayan, M. Aloqaily, and M. Guizani, "Deep federated learning for IoT-based decentralized healthcare systems," in *Proc. Int. Wireless Commun. Mobile Comput. (IWCMC)*, 2021, pp. 105–109.

[11] S. Pandya et al., "Federated learning for smart cities: A comprehensive survey," *Sustain. Energy Technol. Assess.*, vol. 55, Feb. 2023, Art. no. 102987.

[12] S. P. Ramu et al., "Federated learning enabled digital twins for smart cities: Concepts, recent advances, and future directions," *Sustain. Cities Soc.*, vol. 79, Apr. 2022, Art. no. 103663.

[13] S. Savazzi, M. Nicoli, and V. Rampa, "Federated learning with cooperating devices: A consensus approach for massive IoT Networks," *IEEE Internet Things J.*, vol. 7, no. 5, pp. 4641–4654, May 2020.

[14] Z. Du, C. Wu, T. Yoshinaga, K.-L. A. Yau, Y. Ji, and J. Li, "Federated learning for vehicular Internet of Things: Recent advances and open issues," *IEEE Open J. Comput. Soc.*, vol. 1, pp. 45–61, 2020.

[15] L. U. Khan, W. Saad, Z. Han, E. Hossain, and C. S. Hong, "Federated learning for Internet of Things: Recent advances, taxonomy, and open challenges," *IEEE Commun. Surveys Tuts.*, vol. 23, no. 3, pp. 1759–1799, 3rd Quart., 2021.

[16] A. Imteaj, U. Thakker, S. Wang, J. Li, and M. H. Amini, "A survey on federated learning for resource-constrained IoT devices," *IEEE Internet Things J.*, vol. 9, no. 1, pp. 1–24, Jan. 2022.

[17] O. R. A. Almanifi, C.-O. Chow, M.-L. Tham, J. H. Chuah, and J. Kanesan, "Communication and computation efficiency in federated learning: A survey," *Internet Things*, vol. 22, Jul. 2023, Art. no. 100742.

[18] A. Rauniyar et al., "Federated learning for medical applications: A taxonomy, current trends, challenges, and future research directions," *IEEE Internet Things J.*, vol. 11, no. 5, pp. 7374–7398, Mar. 2024.

[19] P. Qi, D. Chiaro, A. Guzzo, M. Ianni, G. Fortino, and F. Piccialli, "Model aggregation techniques in federated learning: A comprehensive survey," *Future Gener. Comput. Syst.*, vol. 150, pp. 272–293, Jan. 2024.

[20] M. H. Alsharif, R. Kannadasan, W. Wei, K. S. Nisar, and A.-H. Abdel-Aty, "A contemporary survey of recent advances in federated learning: Taxonomies, applications, and challenges," *Internet Things*, vol. 27, Oct. 2024, Art. no. 101251.

[21] V. K. Quy, D. C. Nguyen, D. V. Anh, and N. M. Quy, "Federated learning for green and sustainable 6G IIoT applications," *Internet Things*, vol. 25, Apr. 2024, Art. no. 101061.

[22] N. Rieke et al., "The future of digital health with federated learning," *NPJ Digit. Med.*, vol. 3, no. 1, p. 119, 2020.

[23] S. AbdulRahman, H. Tout, H. Ould-Slimane, A. Mourad, C. Talhi, and M. Guizani, "A survey on federated learning: The journey from centralized to distributed on-site learning and beyond," *IEEE Internet Things J.*, vol. 8, no. 7, pp. 5476–5497, Apr. 2021.

[24] L. Li, Y. Fan, M. Tse, and K.-Y. Lin, "A review of applications in federated learning," *Comput. Ind. Eng.*, vol. 149, Nov. 2020, Art. no. 106854.

[25] C. Zhang, Y. Xie, H. Bai, B. Yu, W. Li, and Y. Gao, "A survey on federated learning," *Knowl.-Based Syst.*, vol. 216, Mar. 2021, Art. no. 106775.

[26] D. C. Nguyen, M. Ding, P. N. Pathirana, A. Seneviratne, J. Li, and H. V. Poor, "Federated learning for Internet of Things: A comprehensive survey," *IEEE Commun. Surveys Tuts.*, vol. 23, no. 3, pp. 1622–1658, 3rd Quart., 2021.

[27] A. Imteaj, K. M. Ahmed, U. Thakker, S. Wang, J. Li, and M. H. Amini, "Federated learning for resource-constrained IoT devices: Panoramas and state of the art," *Federated and Transfer Learning* (Adaptation, Learning, and Optimization), vol. 27, R. Razavi-Far, B. Wang, M. E. Taylor, and Q. Yang, Eds., Cham, Switzerland: Springer, 2022, pp. 7–27. [Online]. Available: https://doi.org/10.1007/978-3-031-11748-0_2

[28] W. Y. B. Lim et al., "Federated learning in mobile edge networks: A comprehensive survey," *IEEE Commun. Surveys Tuts.*, vol. 22, no. 3, pp. 2031–2063, 3rd Quart., 2020.

[29] B. McMahan, E. Moore, D. Ramage, S. Hampson, and B. A. y Arcas, "Communication-efficient learning of deep networks from decentralized data," in *Proc. Artif. Intell. Stat.*, 2017, pp. 1273–1282.

[30] Y. Chen, X. Qin, J. Wang, C. Yu, and W. Gao, "FedHealth: A federated transfer learning framework for wearable healthcare," *IEEE Intell. Syst.*, vol. 35, no. 4, pp. 83–93, Jul./Aug. 2020.

[31] X. Zhou et al., "Hierarchical federated learning with social context clustering-based participant selection for Internet of Medical Things applications," *IEEE Trans. Comput. Social Syst.*, vol. 10, no. 4, pp. 1742–1751, Aug. 2023.

[32] B. Yuan, S. Ge, and W. Xing, "A federated learning framework for healthcare IoT devices," 2020, *arXiv:2005.05083*.

[33] L. Huang, A. L. Shea, H. Qian, A. Masurkar, H. Deng, and D. Liu, "Patient clustering improves efficiency of federated machine learning to predict mortality and hospital stay time using distributed electronic medical records," *J. Biomed. Informat.*, vol. 99, Nov. 2019, Art. no. 103291.

[34] S. Lu, Y. Zhang, and Y. Wang, "Decentralized federated learning for electronic health records," in *Proc. 54th Annu. Conf. Inf. Sci. Syst. (CISS)*, 2020, pp. 1–5.

[35] S. Warnat-Herresthal et al., "Swarm learning for decentralized and confidential clinical machine learning," *Nature*, vol. 594, no. 7862, pp. 265–270, 2021.

[36] H. R. Roth et al., "Federated learning for breast density classification: A real-world implementation," in *Proc. 2nd MICCAI Workshop Domain Adaptation Represent. Transfer, Distrib. Collaborative Learn.*, 2020, pp. 181–191.

[37] B. Yuan and J. Li, "The policy effect of the general data protection regulation (GDPR) on the digital public health sector in the European union: an empirical investigation," *Int. J. Environ. Res. Public Health*, vol. 16, no. 6, p. 1070, 2019.

[38] M. M. Salim and J. H. Park, "Federated learning-based secure electronic health record sharing scheme in medical informatics," *IEEE J. Biomed. Health Inform.*, vol. 27, no. 2, pp. 617–624, Feb. 2023.

[39] T. V. Nguyen et al., "A novel decentralized federated learning approach to train on globally distributed, poor quality, and protected private medical data," *Sci. Rep.*, vol. 12, no. 1, p. 8888, 2022.

[40] W. Li et al, "Privacy-preserving federated brain tumour segmentation," in *Proc. 10th Int. Workshop Mach. Learn. Med. Imag.*, Shenzhen, China, 2019, pp. 133–141.

[41] A. N. Giaquinto et al., "Breast cancer statistics 2022," *CA Cancer J. Clin.*, vol. 72, no. 6, pp. 524–541, 2022.

[42] P. Ritchie, *Cause of Death*. Edinburgh, U.K.: Black & White Publ. Ltd., 2017.

[43] M. Humayun, M. I. Khalil, S. N. Almuayqil, and N. Z. Jhanjhi, "Framework for detecting breast cancer risk presence using deep learning," *Electronics*, vol. 12, no. 2, p. 403, 2023.

[44] S. J. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach*. Boston, MA, USA: Pearson, 2016.

[45] B. J. Erickson, P. Korfiatis, Z. Akkus, and T. L. Kline, "Machine learning for medical imaging," *Radiographics*, vol. 37, no. 2, pp. 505–515, 2017.

[46] U. Farooq, N. Tariq, M. Asim, T. Baker, and A. Al-Shamma'a, "Machine learning and the Internet of Things security: Solutions and open challenges," *J. Parallel Distrib. Comput.*, vol. 162, pp. 89–104, Apr. 2022.

[47] A. Qayyum, K. Ahmad, M. A. Ahsan, A. Al-Fuqaha, and J. Qadir, "Collaborative federated learning for healthcare: Multi-modal Covid-19 diagnosis at the edge," *IEEE Open J. Comput. Soc.*, vol. 3, pp. 172–184, 2022.

[48] H. Razzaghi, M. A. Troester, G. L. Gierach, A. F. Olshan, B. C. Yankaskas, and R. C. Millikan, "Mammographic density and breast cancer risk in white and African American women," *Breast Cancer Res. Treatment*, vol. 135, pp. 571–580, Aug. 2012.

[49] A. Albaseer, B. S. Ciftler, M. Abdallah, and A. Al-Fuqaha, "Exploiting unlabeled data in smart cities using federated edge learning," in *Proc. Int. Wireless Commun. Mobile Comput. (IWCMC)*, 2020, pp. 1666–1671.

[50] Y. Liu, J. Q. James, J. Kang, D. Niyato, and S. Zhang, "Privacy-preserving traffic flow prediction: A federated learning approach," *IEEE Internet Things J.*, vol. 7, no. 8, pp. 7751–7763, Aug. 2020.

[51] G. Hua, L. Zhu, J. Wu, C. Shen, L. Zhou, and Q. Lin, "Blockchain-based federated learning for intelligent control in heavy haul railway," *IEEE Access*, vol. 8, pp. 176830–176839, 2020.

[52] Z. Yu, J. Hu, G. Min, Z. Zhao, W. Miao, and M. S. Hossain, "Mobility-aware proactive edge caching for connected vehicles using federated learning," *IEEE Trans. Intell. Transp. Syst.*, vol. 22, no. 8, pp. 5341–5351, Aug. 2021.

[53] Q. Chen, Z. You, and H. Jiang, "Semi-asynchronous hierarchical federated learning for cooperative intelligent transportation systems," 2021, *arXiv:2110.09073*.

[54] W. Yamany, N. Moustafa, and B. Turnbull, "OQFL: An optimized quantum-based federated learning framework for defending against adversarial attacks in intelligent transportation systems," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 1, pp. 893–903, Jan. 2023.

[55] A. M. Elbir, B. Soner, S. Çöleri, D. Gündüz, and M. Bennis, "Federated learning in vehicular networks," in *Proc. IEEE Int. Mediterr. Conf. Commun. Netw. (MeditCom)*, 2022, pp. 72–77.

[56] C. Zhang, S. Zhang, J. Q. James, and S. Yu, "FastGNN: A topological information protected federated learning approach for traffic speed forecasting," *IEEE Trans. Ind. Informat.*, vol. 17, no. 12, pp. 8464–8474, Dec. 2021.

[57] D. M. Manias and A. Shami, "Making a case for federated learning in the Internet of Vehicles and intelligent transportation systems," *IEEE Netw.*, vol. 35, no. 3, pp. 88–94, May/Jun. 2021.

[58] A. Zanella, N. Bui, A. Castellani, L. Vangelista, and M. Zorzi, "Internet of Things for smart cities," *IEEE Internet Things J.*, vol. 1, no. 1, pp. 22–32, Feb. 2014.

[59] W. Jiang and J. Luo, "Graph neural network for traffic forecasting: A survey," *Expert Syst. Appl.*, vol. 207, Nov. 2022, Art. no. 117921.

[60] M. M. Hamed, H. R. Al-Masaeid, and Z. M. B. Said, "Short-term prediction of traffic volume in urban arterials," *J. Transport. Eng.*, vol. 121, no. 3, pp. 249–254, 1995.

[61] X. Liang, Y. Liu, T. Chen, M. Liu, and Q. Yang, "Federated transfer reinforcement learning for autonomous driving," in *Federated and Transfer Learning*. Cham, Switzerland: Springer, 2022, pp. 357–371.

[62] H. Jiang, M. Liu, B. Yang, Q. Liu, J. Li, and X. Guo, "Customized federated learning for accelerated edge computing with heterogeneous task targets," *Comput. Netw.*, vol. 183, Dec. 2020, Art. no. 107569.

[63] Z. Yu, J. Hu, G. Min, H. Xu, and J. Mills, "Proactive content caching for Internet-of-Vehicles based on peer-to-peer federated learning," in *Proc. IEEE 26th Int. Conf. Parallel Distrib. Syst. (ICPADS)*, 2020, pp. 601–608.

[64] L. Barbieri, S. Savazzi, M. Brambilla, and M. Nicoli, "Decentralized federated learning for extended sensing in 6G connected vehicles," *Veh. Commun.*, vol. 33, Jan. 2022, Art. no. 100396.

[65] J.-H. Chen, M.-R. Chen, G.-Q. Zeng, and J.-S. Weng, "BDFL: A Byzantine-fault-tolerance decentralized federated learning method for autonomous vehicle," *IEEE Trans. Veh. Technol.*, vol. 70, no. 9, pp. 8639–8652, Sep. 2021.

[66] A. Hammoud, H. Otrok, A. Mourad, and Z. Dziong, "On demand fog federations for horizontal federated learning in IoV," *IEEE Trans. Netw. Service Manag.*, vol. 19, no. 3, pp. 3062–3075, Sep. 2022.

[67] C. Chen, "Freeway performance measurement system (PeMS)," Ph.D. dissertation, Dept. Transport., Univ. California, Berkeley, CA, USA, 2002.

[68] Al-H. Rasha, T. Li, W. Huang, J. Gu, and C. Li, "Federated learning in smart cities: Privacy and security survey," *Inf. Sci.*, vol. 632, pp. 833–857, Jun. 2023.

[69] Y. Lu, X. Huang, K. Zhang, S. Maharjan, and Y. Zhang, "Blockchain empowered asynchronous federated learning for secure data sharing in Internet of Vehicles," *IEEE Trans. Veh. Technol.*, vol. 69, no. 4, pp. 4298–4311, Apr. 2020.

[70] S. R. Pokhrel and J. Choi, "Federated learning with blockchain for autonomous vehicles: Analysis and design challenges," *IEEE Trans. Commun.*, vol. 68, no. 8, pp. 4734–4746, Aug. 2020.

[71] F. Bragato et al., "Towards decentralized predictive quality of service in next-generation vehicular networks," 2023, *arXiv:2302.11268*.

[72] X. Hu, R. Li, L. Wang, Y. Ning, and K. Ota, "A data sharing scheme based on federated learning in IoV," *IEEE Trans. Veh. Technol.*, vol. 72, no. 9, pp. 11644–11656, Sep. 2023.

[73] Z. M. Kassas, P. Closas, and J. Gross, "Navigation systems panel report navigation systems for autonomous and semi-autonomous vehicles: Current trends and future challenges," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 34, no. 5, pp. 82–84, May 2019.

[74] M. Alqarqaz, M. B. Younes, and R. Qaddoura, "An object classification approach for autonomous vehicles using machine learning techniques," *World Electr. Veh. J.*, vol. 14, no. 2, p. 41, 2023.

[75] M. Hataba, A. Sherif, M. Mahmoud, M. Abdallah, and W. Alasmary, "Security and privacy issues in autonomous vehicles: A layer-based survey," *IEEE Open J. Commun. Soc.*, vol. 3, pp. 811–829, 2022.

[76] S. Behera, M. Adhikari, V. G. Menon, and M. A. Khan, "Large model-assisted federated learning for object detection of autonomous vehicles in edge," *IEEE Trans. Veh. Technol.*, early access, May 22, 2024, doi: 10.1109/TVT.2024.3404097.

[77] A. Geiger, P. Lenz, and R. Urtasun, "Are we ready for autonomous driving? The KITTI vision benchmark suite," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2012, pp. 3354–3361.

[78] A. Hard et al., "Federated learning for mobile keyboard prediction," 2018, *arXiv:1811.03604*.

[79] T. S. Brisimi, R. Chen, T. Mela, A. Olshevsky, I. C. Paschalidis, and W. Shi, "Federated learning of predictive models from federated electronic health records," *Int. J. Med. Inform.*, vol. 112, pp. 59–67, Apr. 2018.

[80] S. Lu, Y. Yao, and W. Shi, "Collaborative learning on the edges: A case study on connected vehicles," in *Proc. 2nd USENIX Workshop Hot Topics Edge Comput. (HotEdge)*, 2019, pp. 1–8.

[81] S. Caldas, J. Konečny, H. B. McMahan, and A. Talwalkar, "Expanding the reach of federated learning by reducing client resource requirements," 2018, *arXiv:1812.07210*.

[82] J. Konečnỳ, H. B. McMahan, F. X. Yu, P. Richtárik, A. T. Suresh, and D. Bacon, "Federated learning: Strategies for improving communication efficiency," 2016, *arXiv:1610.05492*.

[83] Z. Xu, Z. Yang, J. Xiong, J. Yang, and X. Chen, "ELFISH: Resource-aware federated learning on heterogeneous edge devices," *Ratio*, vol. 2, pp. 1–6, Dec. 2019.

[84] S. Wang et al., "Adaptive federated learning in resource constrained edge computing systems," *IEEE J. Sel. Areas Commun.*, vol. 37, no. 6, pp. 1205–1221, Jun. 2019.

[85] X. Wang, Y. Han, C. Wang, Q. Zhao, X. Chen, and M. Chen, "In-edge AI: Intelligentizing mobile edge computing, caching and communication by federated learning," *IEEE Netw.*, vol. 33, no. 5, pp. 156–165, Sep./Oct. 2019.

[86] J. Mills, J. Hu, and G. Min, "Communication-efficient federated learning for wireless edge intelligence in IoT," *IEEE Internet Things J.*, vol. 7, no. 7, pp. 5986–5994, Jul. 2020.

[87] I. Nemer, T. Sheltami, E. Shakshuki, A. A. Elkhail, and M. Adam, "Performance evaluation of range-free localization algorithms for wireless sensor networks," *Personal Ubiquitous Comput.*, vol. 25, no. 1, pp. 177–203, May 2020.

[88] S. Kaleem, A. Sohail, M. Babar, A. Ahmad, and M. U. Tariq, "A hybrid model for energy-efficient green Internet of Things enabled intelligent transportation systems using federated learning." *Internet Things*, vol. 25, Apr. 2024, Art. no. 101038.

[89] N. H. Tran, W. Bao, A. Zomaya, M. N. Nguyen, and C. S. Hong, "Federated learning over wireless networks: Optimization model design and analysis," in *Proc. IEEE Conf. Comput. Commun.*, 2019, pp. 1387–1395.

[90] H. Kim, J. Park, M. Bennis, and S.-L. Kim, "Blockchained on-device federated learning," *IEEE Commun. Lett.*, vol. 24, no. 6, pp. 1279–1283, Jun. 2020.

[91] L. Liu, J. Zhang, S. H. Song, and K. B. Letaief, "Client-edge-cloud hierarchical federated learning," in *Proc. IEEE Int. Conf. Commun. (ICC)*, 2020, pp. 1–6.

[92] M. S. H. Abad, E. Ozfatura, D. Gunduz, and O. Ercetin, "Hierarchical federated learning across heterogeneous cellular networks," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, 2020, pp. 8866–8870.

[93] S. M. Rajagopal, M. Supriya, and R. Buyya, "FedSDM: Federated learning based smart decision making module for ECG data in IoT integrated edge-fog-cloud computing environments," *Internet Things*, vol. 22, Jul. 2023, Art. no. 100784.

[94] R. Saha, S. Misra, and P. K. Deb, "FogFL: Fog-assisted federated learning for resource-constrained IoT devices," *IEEE Internet Things J.*, vol. 8, no. 10, pp. 8456–8463, May 2021.

[95] E. T. M. Beltrán et al., "Decentralized federated learning: Fundamentals, state of the art, frameworks, trends, and challenges," *IEEE Commun. Surveys Tuts.*, vol. 25, no. 4, pp. 2983–3013, 4th Quart., 2023.

[96] H. Zhang, J. Bosch, and H. H. Olsson, "Federated learning systems: Architecture alternatives," in *Proc. 27th Asia-Pacific Softw. Eng. Conf. (APSEC)*, 2020, pp. 385–394.

[97] O. Rana et al., "Hierarchical and decentralised federated learning," in *Proc. Cloud Continuum*, 2022, pp. 1–9.

[98] X. Chen, G. Xu, X. Xu, H. Jiang, Z. Tian, and T. Ma, "Multicenter hierarchical federated learning with fault-tolerance mechanisms for resilient edge computing networks," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Mar. 28, 2024, doi: 10.1109/TNNLS.2024.3362974.

[99] L. Witt, M. Heyer, K. Toyoda, W. Samek, and D. Li, "Decentral and incentivized federated learning frameworks: A systematic literature review," *IEEE Internet Things J.*, vol. 10, no. 4, pp. 3642–3663, Feb. 2023.

[100] I. Hegedűs, G. Danner, and M. Jelasity, "Gossip learning as a decentralized alternative to federated learning," in *Proc. 19th IFIP WG 6.1 Int. Conf. Distrib. Appl. Interoperable Syst.*, 2019, pp. 74–90.

[101] R. Xu and Y. Chen, "μDFL: A secure microchained decentralized federated learning fabric atop IoT networks," *IEEE Trans. Netw. Service Manag.*, vol. 19, no. 3, pp. 2677–2688, Sep. 2022.

[102] X. Yin, Y. Zhu, and J. Hu, "A comprehensive survey of privacy-preserving federated learning: A taxonomy, review, and future directions," *ACM Comput. Surveys*, vol. 54, no. 6, pp. 1–36, 2021.

[103] V. Mothukuri, R. M. Parizi, S. Pouriyeh, Y. Huang, A. Dehghantanha, and G. Srivastava, "A survey on security and privacy of federated learning," *Future Gener. Comput. Syst.*, vol. 115, pp. 619–640, Feb. 2021.

[104] Q. Yang, Y. Liu, T. Chen, and Y. Tong, "Federated machine learning: Concept and applications," *ACM Trans. Intell. Syst. Technol.*, vol. 10, no. 2, pp. 1–19, 2019.

[105] S. I. Manzoor, S. Jain, Y. Singh, and H. Singh, "Federated learning based privacy ensured sensor communication in IoT networks: A taxonomy, threats and attacks," *IEEE Access*, vol. 11, pp. 42248–42275, 2023.

[106] T. Yang et al., "Applied federated learning: Improving Google keyboard query suggestions," 2018, *arXiv:1812.02903*.

[107] M. H. Mahmoud, A. Albaseer, M. Abdallah, and N. Al-Dhahir, "Federated learning resource optimization and client selection for total energy minimization under outage, latency, and bandwidth constraints with partial or no CSI," *IEEE Open J. Commun. Soc.*, vol. 4, pp. 936–953, 2023.

[108] J. Kang, Z. Xiong, D. Niyato, S. Xie, and J. Zhang, "Incentive mechanism for reliable federated learning: A joint optimization approach to combining reputation and contract theory," *IEEE Internet Things J.*, vol. 6, no. 6, pp. 10700–10714, Dec. 2019.

[109] P. Kairouz et al., "Advances and open problems in federated learning," *Found. Trends® Mach. Learn.*, vol. 14, nos. 1–2, pp. 1–210, 2021.

[110] O. Marfoq, C. Xu, G. Neglia, and R. Vidal, "Throughput-optimal topology design for cross-silo federated learning," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 19478–19487.

[111] K. Cheng et al., "SecureBoost: A lossless federated learning framework," *IEEE Intell. Syst.*, vol. 36, no. 6, pp. 87–98, Nov./Dec. 2021.

[112] Q. Li, Z. Wen, and B. He, "Practical federated gradient boosting decision trees," in *Proc. AAAI Conf. Artif. Intell.*, vol. 34, 2020, pp. 4642–4649.

[113] S. Hardy et al., "Private federated learning on vertically partitioned data via entity resolution and additively homomorphic encryption," 2017, *arXiv:1711.10677*.

[114] V. Nikolaenko, U. Weinsberg, S. Ioannidis, M. Joye, D. Boneh, and N. Taft, "Privacy-preserving ridge regression on hundreds of millions of records," in *Proc. IEEE Symp. Security Privacy*, 2013, pp. 334–348.

[115] T. Li, A. K. Sahu, A. Talwalkar, and V. Smith, "Federated learning: Challenges, methods, and future directions," *IEEE Signal Process. Mag.*, vol. 37, no. 3, pp. 50–60, May 2020.

[116] H. Wang, M. Yurochkin, Y. Sun, D. Papailiopoulos, and Y. Khazaeni, "Federated learning with matched averaging," 2020, *arXiv:2002.06440*.

[117] R. Shokri and V. Shmatikov, "Privacy-preserving deep learning," in *Proc. 22nd ACM SIGSAC Conf. Comput. Commun. Security*, 2015, pp. 1310–1321.

[118] M. H. Shirvani and M. Masdari, "A survey study on trust-based security in Internet of Things: Challenges and issues," *Internet Things*, vol. 21, Apr. 2023, Art. no. 100640.

[119] H. Lee, J. Kim, S. Ahn, R. Hussain, S. Cho, and J. Son, "Digestive neural networks: A novel defense strategy against inference attacks in federated learning," *Comput. Security*, vol. 109, Oct. 2021, Art. no. 102378.

[120] L. T. Phong, Y. Aono, T. Hayashi, L. Wang, and S. Moriai, "Privacy-preserving deep learning via additively homomorphic encryption," *IEEE Trans. Inf. Forensics Security*, vol. 13, no. 5, pp. 1333–1345, 2017.

[121] F. Haddadpour, M. M. Kamani, M. Mahdavi, and V. Cadambe, "Trading redundancy for communication: Speeding up distributed SGD for non-convex optimization," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 2545–2554.

[122] A. Nilsson, S. Smith, G. Ulm, E. Gustavsson, and M. Jirstrand, "A performance evaluation of federated learning algorithms," in *Proc. 2nd Workshop Distrib. Infrastruct. Deep Learn.*, 2018, pp. 1–8.

[123] V. Smith, C.-K. Chiang, M. Sanjabi, and A. S. Talwalkar, "Federated multi-task learning," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 30, 2017, pp. 1–14.

[124] X. Yi, R. Li, C. Peng, J. Wu, and Z. Zhao, "HFedMTL: Hierarchical federated multi-task learning," in *Proc. IEEE 33rd Annu. Int. Symp. Personal, Indoor Mobile Radio Commun. (PIMRC)*, 2022, pp. 1–6.

[125] A. Li, J. Sun, X. Zeng, M. Zhang, H. Li, and Y. Chen, "FedMask: Joint computation and communication-efficient personalized federated learning via heterogeneous masking," in *Proc. 19th ACM Conf. Embedded Netw. Sens. Syst.*, 2021, pp. 42–55.

[126] B. Luo, X. Li, S. Wang, J. Huang, and L. Tassiulas, "Cost-effective federated learning design," in *Proc. IEEE Conf. Comput. Commun.*, 2021, pp. 1–10.

[127] Y. Jiang et al., "Model pruning enables efficient federated learning on edge devices," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 12, pp. 10374–10386, Dec. 2023.

[128] A. Das and T. Brunschwiler, "Privacy is what we care about: Experimental investigation of federated learning on edge devices," in *Proc. 1st Int. Workshop Challenges Artif. Intell. Mach. Learn. Internet Things*, 2019, pp. 39–42.

[129] S. Zheng, C. Shen, and X. Chen, "Design and analysis of uplink and downlink communications for federated learning," *IEEE J. Sel. Areas Commun.*, vol. 39, no. 7, pp. 2150–2167, Jul. 2021.

[130] Y. Shi, K. Yang, T. Jiang, J. Zhang, and K. B. Letaief, "Communication-efficient edge AI: Algorithms and systems," *IEEE Commun. Surveys Tuts.*, vol. 22, no. 4, pp. 2167–2191, 4th Quart., 2020.

[131] J. Hamer, M. Mohri, and A. T. Suresh, "FedBoost: A communication-efficient algorithm for federated learning," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 3973–3983.

[132] X. Yao, T. Huang, C. Wu, R.-X. Zhang, and L. Sun, "Federated learning with additional mechanisms on clients to reduce communication costs," 2019, *arXiv:1908.05891*.

[133] N. Srivastava, G. Hinton, A. Krizhevsky, I. Sutskever, and R. Salakhutdinov, "Dropout: A simple way to prevent neural networks from overfitting," *J. Mach. Learn. Res.*, vol. 15, no. 1, pp. 1929–1958, 2014.

[134] T. Li, M. Sanjabi, A. Beirami, and V. Smith, "Fair resource allocation in federated learning," 2019, *arXiv:1905.10497*.

[135] L. Lyu, X. Xu, Q. Wang, and H. Yu, "Collaborative fairness in federated learning," in *Federated Learning: Privacy and Incentive* (Lecture Notes in Computer Science 12500). Cham, Switzerland: Springer, 2020, pp. 189–204. [Online]. Available: https://doi.org/10.1007/978-3-030-63076-8_14

[136] H. Yu et al., "A fairness-aware incentive scheme for federated learning," in *Proc. AAAI/ACM Conf. AI, Ethics, Soc.*, 2020, pp. 393–399.

[137] H. Gao, A. Xu, and H. Huang, "On the convergence of communication-efficient local SGD for federated learning," in *Proc. AAAI Conf. Artif. Intell.*, vol. 35, 2021, pp. 7510–7518.

[138] S. Bibikar, H. Vikalo, Z. Wang, and X. Chen, "Federated dynamic sparse training: Computing less, communicating less, yet learning better," in *Proc. AAAI Conf. Artif. Intell.*, vol. 36, 2022, pp. 6080–6088.

[139] H.-P. Wang, S. Stich, Y. He, and M. Fritz, "ProgFED: effective, communication, and computation efficient federated learning by progressive training," in *Proc. Int. Conf. Mach. Learn.*, 2022, pp. 23034–23054.

[140] W. Luping, W. Wei, and L. Bo, "CMFL: Mitigating communication overhead for federated learning," in *Proc. IEEE 39th Int. Conf. Distrib. Comput. Syst. (ICDCS)*, 2019, pp. 954–964.

[141] Y. Xiong, R. Wang, M. Cheng, F. Yu, and C.-J. Hsieh, "FedDM: Iterative distribution matching for communication-efficient federated learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 16323–16332.

[142] R. Dai, L. Shen, F. He, X. Tian, and D. Tao, "DisPFL: Towards communication-efficient personalized federated learning via decentralized sparse training," 2022, *arXiv:2206.00187*.

[143] C. Ma et al., "Distributed optimization with arbitrary local solvers," *Optim. Methods Softw.*, vol. 32, no. 4, pp. 813–848, 2017.

[144] W. Huang, T. Li, D. Wang, S. Du, and J. Zhang, "Fairness and accuracy in federated learning," 2020, *arXiv:2012.10069*.

[145] J. Hong, H. Wang, Z. Wang, and J. Zhou, "Efficient split-mix federated learning for on-demand and in-situ customization," 2022, *arXiv:2203.09747*.

[146] M. Mohri, G. Sivek, and A. T. Suresh, "Agnostic federated learning," in *Proc. Int. Conf. Mach. Learn.*, 2019, pp. 4615–4625.

[147] K. Pfeiffer, M. Rapp, R. Khalili, and J. Henkel, "Federated learning for computationally-constrained heterogeneous devices: A survey," *ACM Comput. Surveys*, vol. 55, no. 14, pp. 1–27, 2023.

[148] J. Kim, G. Kim, and B. Han, "Multi-level branched regularization for federated learning," in *Proc. Int. Conf. Mach. Learn.*, 2022, pp. 11058–11073.

[149] J. Kirkpatrick et al., "Overcoming catastrophic forgetting in neural networks," *Proc. Nat. Acad. Sci.*, vol. 114, no. 13, pp. 3521–3526, 2017.

[150] S. P. Karimireddy, S. Kale, M. Mohri, S. Reddi, S. Stich, and A. T. Suresh, "SCAFFOLD: Stochastic controlled averaging for federated learning," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 5132–5143.

[151] Q. Li, B. He, and D. Song, "Model-contrastive federated learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 10713–10722.

[152] Y. Laguel, K. Pillutla, J. Malick, and Z. Harchaoui, "Device heterogeneity in federated learning: A superquantile approach," 2020, *arXiv:2002.11223*.

[153] W. Huang, M. Ye, Z. Shi, H. Li, and B. Du, "Rethinking federated learning with domain shift: A prototype view," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2023, pp. 16312–16322.

[154] D. Chen, J. Hu, V. J. Tan, X. Wei, and E. Wu, "Elastic aggregation for federated optimization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 12187–12197.

[155] A. Ghosh, J. Chung, D. Yin, and K. Ramchandran, "An efficient framework for clustered federated learning," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 19586–19597.

[156] Y. Ruan and C. Joe-Wong, "FedSOFT: Soft clustered federated learning with proximal local updating," in *Proc. AAAI Conf. Artif. Intell.*, vol. 36, 2022, pp. 8124–8131.

[157] D. K. Dennis, T. Li, and V. Smith, "Heterogeneity for the win: One-shot federated clustering," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 2611–2620.

[158] H. Wang, Z. Kaplan, D. Niu, and B. Li, "Optimizing federated learning on non-iid data with reinforcement learning," in *Proc. IEEE Conf. Comput. Commun.*, 2020, pp. 1698–1707.

[159] G. Hinton, O. Vinyals, and J. Dean, "Distilling the knowledge in a neural network," 2015, *arXiv:1503.02531*.

[160] C. Briggs, Z. Fan, and P. Andras, "Federated learning with hierarchical clustering of local updates to improve training on non-iid data," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, 2020, pp. 1–9.

[161] F. Sattler, K.-R. Müller, and W. Samek, "Clustered federated learning: Model-agnostic distributed multitask optimization under privacy constraints," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 32, no. 8, pp. 3710–3722, Aug. 2021.

[162] L. Fu, H. Zhang, G. Gao, M. Zhang, and X. Liu, "Client selection in federated learning: Principles, challenges, and opportunities," *IEEE Internet Things J.*, vol. 10, no. 24, pp. 21811–21819, Dec. 2023.

[163] M. Yang, X. Wang, H. Zhu, H. Wang, and H. Qian, "Federated learning with class imbalance reduction," in *Proc. 29th Eur. Signal Process. Conf. (EUSIPCO)*, 2021, pp. 2174–2178.

[164] M. Tang et al., "FedCor: Correlation-based active client selection strategy for heterogeneous federated learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 10102–10111.

[165] Z. Qin, L. Yang, Q. Wang, Y. Han, and Q. Hu, "Reliable and interpretable personalized federated learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 20422–20431.

[166] D. Makhija, X. Han, N. Ho, and J. Ghosh, "Architecture agnostic federated learning for neural networks," in *Proc. Int. Conf. Mach. Learn.*, 2022, pp. 14860–14870.

[167] Q. Wu, K. He, and X. Chen, "Personalized federated learning for intelligent IoT applications: A cloud-edge based framework," *IEEE Open J. Comput. Soc.*, vol. 1, pp. 35–44, 2020.

[168] T. Lin, L. Kong, S. U. Stich, and M. Jaggi, "Ensemble distillation for robust model fusion in federated learning," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 2351–2363.

[169] S. Itahara, T. Nishio, Y. Koda, M. Morikura, and K. Yamamoto, "Distillation-based semi-supervised federated learning for communication-efficient collaborative training with non-iid private data," *IEEE Trans. Mobile Comput.*, vol. 22, no. 1, pp. 191–205, Jan. 2023.

[170] C. He, M. Annavaram, and S. Avestimehr, "Group knowledge transfer: Federated learning of large CNNs at the edge," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 14068–14080.

[171] Y. J. Cho, A. Manoel, G. Joshi, R. Sim, and D. Dimitriadis, "Heterogeneous ensemble knowledge transfer for training large models in federated learning," 2022, *arXiv:2204.12703*.

[172] S. Stanton, P. Izmailov, P. Kirichenko, A. A. Alemi, and A. G. Wilson, "Does knowledge distillation really work?" in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 6906–6919.

[173] H. Wang et al., "Attack of the tails: Yes, you really can backdoor federated learning," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 3, 2020, pp. 16070–16084.

[174] G. Cheng, Z. Charles, Z. Garrett, and K. Rush, "Does federated dropout actually work?" in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 3387–3395.

[175] E. Diao, J. Ding, and V. Tarokh, "HeteroFL: Computation and communication efficient federated learning for heterogeneous clients," 2020, *arXiv:2010.01264*.

[176] S. Horvath, S. Laskaridis, M. Almeida, I. Leontiadis, S. Venieris, and N. Lane, "FjORD: Fair and accurate federated learning under heterogeneous targets with ordered dropout," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 12876–12889.

[177] Z. Zhang et al., "Refiner: A reliable incentive-driven federated learning system powered by blockchain," *Proc. VLDB Endow.*, vol. 14, no. 12, pp. 2659–2662, 2021.

[178] H. B. Desai, M. S. Ozdayi, and M. Kantarcioglu, "BlockFLA: Accountable federated learning via hybrid blockchain architecture," in *Proc. 11th ACM Conf. Data Appl. Security Privacy*, 2021, pp. 101–112.

[179] S. Rahmadika and K.-H. Rhee, "Reliable collaborative learning with commensurate incentive schemes," in *Proc. IEEE Int. Conf. Blockchain (Blockchain)*, 2020, pp. 496–502.

[180] D. Yin, Y. Chen, R. Kannan, and P. Bartlett, "Byzantine-robust distributed learning: Towards optimal statistical rates," in *Proc. Int. Conf. Mach. Learn.*, 2018, pp. 5650–5659.

[181] C. Xie, O. Koyejo, and I. Gupta, "Generalized Byzantine-tolerant SGD," 2018, *arXiv:1802.10116*.

[182] X. Cao, M. Fang, J. Liu, and N. Z. Gong, "FLTrust: Byzantine-robust federated learning via trust bootstrapping," 2020, *arXiv:2012.13995*.

[183] V. Shejwalkar and A. Houmansadr, "Manipulating the Byzantine: Optimizing model poisoning attacks and defenses for federated learning," in *Proc. NDSS*, 2021, pp. 1–14.

[184] J. An and S. Cho, "Variational autoencoder based anomaly detection using reconstruction probability," Dept. Comput. Sci. Eng., Seoul Nat. Univ. Data Min. Center, Seoul, Republic of Korea, Rep. 2015-2, 2015.

[185] J. Lin, M. Du, and J. Liu, "Free-riders in federated learning: Attacks and defenses," 2019, *arXiv:1911.12560*.

[186] Y. Wang, J. Zhou, G. Feng, X. Niu, and S. Qin, "Blockchain assisted federated learning for enabling network edge intelligence," *IEEE Netw.*, vol. 37, no. 1, pp. 96–102, Jan./Feb. 2023.

[187] V. Mugunthan, R. Rahman, and L. Kagal, "BlockFlow: Decentralized, privacy-preserving, and accountable federated machine learning," in *Proc. 3rd Int. Congr. Blockchain Appl.*, 2022, pp. 233–242.

[188] Y. Zhao et al., "Privacy-preserving blockchain-based federated learning for IoT devices," *IEEE Internet Things J.*, vol. 8, no. 3, pp. 1817–1829, Feb. 2021.

[189] S. Kumar, S. Dutta, S. Chatturvedi, and M. P. S. Bhatia, "Strategies for enhancing training and privacy in blockchain enabled federated learning," in *Proc. IEEE 6th Int. Conf. Multimedia Big Data (BigMM)*, 2020, pp. 333–340.

[190] B. Wang, Y. Chen, H. Jiang, and Z. Zhao, "PPeFL: Privacy-preserving edge federated learning with local differential privacy," *IEEE Internet Things J.*, vol. 10, no. 17, pp. 15488–15500, Sep. 2023.

[191] Y. Zhao, M. Li, L. Lai, N. Suda, D. Civin, and V. Chandra, "Federated learning with non-iid data," 2018, *arXiv:1806.00582*.

[192] X. Qu, S. Wang, Q. Hu, and X. Cheng, "Proof of federated learning: A novel energy-recycling consensus algorithm," *IEEE Trans. Parallel Distrib. Syst.*, vol. 32, no. 8, pp. 2074–2085, Aug. 2021.

[193] X. Bao, C. Su, Y. Xiong, W. Huang, and Y. Hu, "FLChain: A blockchain for auditable federated learning with trust and incentive," in *Proc. 5th Int. Conf. Big Data Comput. Commun. (BIGCOM)*, 2019, pp. 151–159.

[194] A. Li, L. Zhang, J. Tan, Y. Qin, J. Wang, and X.-Y. Li, "Sample-level data selection for federated learning," in *Proc. IEEE Conf. Comput. Commun.*, 2021, pp. 1–10.

[195] L. Chen, H. Wang, Z. Charles, and D. Papailiopoulos, "DRACO: Byzantine-resilient distributed training via redundant gradients," in *Proc. Int. Conf. Mach. Learn.*, 2018, pp. 903–912.

[196] R. Bommasani et al., "On the opportunities and risks of foundation models," 2021, *arXiv:2108.07258*.

[197] S. Bubeck et al., "Sparks of artificial general intelligence: Early experiments with GPT-4," 2023, *arXiv:2303.12712*.

[198] Z. Zhang et al., "FEDLEGAL: The first real-world federated learning benchmark for legal NLP," in *Proc. 61st Annu. Meeting Assoc. Comput. Linguist.*, vol. 1, 2023, pp. 3492–3507.

[199] A. Radford, K. Narasimhan, T. Salimans, and I. Sutskever, "Improving language understanding by generative pre-training," in *Proc. OpenAI Blog*, 2018, pp. 1–12.

[200] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," 2018, *arXiv:1810.04805*.

[201] K. Han, A. Xiao, E. Wu, J. Guo, C. Xu, and Y. Wang, "Transformer in transformer," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 15908–15919.

[202] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[203] H. Touvron, M. Cord, M. Douze, F. Massa, A. Sablayrolles, and H. Jégou, "Training data-efficient image transformers & distillation through attention," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 10347–10357.

[204] J. Beal, E. Kim, E. Tzeng, D. H. Park, A. Zhai, and D. Kislyuk, "Toward transformer-based object detection," 2020, *arXiv:2012.09958*.

[205] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2021, pp. 10012–10022.

[206] X. Chen, S. Xie, and K. He, "An empirical study of training self-supervised vision transformers," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2021, pp. 9640–9649.

[207] M. Caron et al., "Emerging properties in self-supervised vision transformers," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2021, pp. 9650–9660.

[208] H. Bao, L. Dong, S. Piao, and F. Wei, "BEiT: BERT pre-training of image transformers," 2021, *arXiv:2106.08254*.

[209] S. Paul and P. Y. Chen, "Vision transformers are robust learners," in *Proc. AAAI Conf. Artif. Intell.*, vol. 36, 2022, pp. 2071–2081.

[210] M. Raghu, T. Unterthiner, S. Kornblith, C. Zhang, and A. Dosovitskiy, "Do vision transformers see like convolutional neural networks?" in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 12116–12128.

[211] S. Cao, P. Xu, and D. A. Clifton, "How to understand masked autoencoders," 2022, *arXiv:2202.03670*.

[212] Z. Xu et al., "Federated learning of Gboard language models with differential privacy," 2023, *arXiv:2305.18465*.

[213] Y. Tian, Y. Wan, L. Lyu, D. Yao, H. Jin, and L. Sun, "FedBERT: When federated learning meets pre-training," *ACM Trans. Intell. Syst. Technol.*, vol. 13, no. 4, pp. 1–26, 2022.

[214] J. Zhang et al., "Towards building the federated GPT: Federated instruction tuning," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, 2024, pp. 6915–6919.

[215] F. Jiang et al., "Personalized wireless federated learning for large language models," 2024, *arXiv:2404.13238*.

[216] L. Qu et al., "Rethinking architecture design for tackling data heterogeneity in federated learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 10061–10071.

[217] G. Sun, M. Mendieta, J. Luo, S. Wu, and C. Chen, "FedPerfix: Towards partial model personalization of vision transformers in federated learning," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2023, pp. 4988–4998.

[218] H. Li et al., "FedTP: Federated learning by transformer personalization," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 10, pp. 13426–13440, Oct. 2024.

[219] H.-Y. Chen, C.-H. Tu, Z. Li, H.-W. Shen, and W.-L. Chao, "On the importance and applicability of pre-training for federated learning," 2022, *arXiv:2206.11488*.

[220] J. Nguyen, J. Wang, K. Malik, M. Sanjabi, and M. Rabbat, "Where to begin? On the impact of pre-training and initialization in federated learning," 2022, *arXiv:2206.15387*.

[221] G. Sun, M. Mendieta, T. Yang, and C. Chen, "Conquering the communication constraints to enable large pre-trained models in federated learning," 2022, *arXiv:2210.01708*.

[222] Z. Zhang et al., "FedPETuning: When federated learning meets the parameter-efficient tuning methods of pre-trained language models," in *Proc. Annu. Meeting Assoc. Comput. Linguist.*, 2023, pp. 9963–9977.

[223] S. Yu, J. P. Muñoz, and A. Jannesari, "Bridging the gap between foundation models and heterogeneous federated learning," 2023, *arXiv:2310.00247*.

[224] W. Lu, X. Hu, J. Wang, and X. Xie, "FedCLIP: Fast generalization and personalization for clip in federated learning," 2023, *arXiv:2302.13485*.

[225] Z. Lit, S. Sit, J. Wang, and J. Xiao, "Federated split bert for heterogeneous text classification," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, 2022, pp. 1–8.

[226] H. Feng, T. Pang, C. Du, W. Chen, S. Yan, and M. Lin, "Does federated learning really need backpropagation?" 2023, *arXiv:2301.12195*.

[227] M. A. Albreem, A. M. Sheikh, M. H. Alsharif, M. Jusoh, and M. N. Mohd Yasin, "Green Internet of Things (GIoT): Applications, practices, awareness, and challenges," *IEEE Access*, vol. 9, pp. 38833–38858, 2021.

[228] Z. H. Ali and H. A. Ali, "Towards sustainable smart IoT applications architectural elements and design: Opportunities, challenges, and open directions," *J. Supercomput.*, vol. 77, no. 6, pp. 5668–5725, 2021.

[229] F. A. Almalki et al., "Green IoT for eco-friendly and sustainable smart cities: Future directions and opportunities," *Mobile Netw. Appl.*, vol. 28, no. 1, pp. 178–202, 2023.

[230] P. Fraga-Lamas, S. I. Lopes, and T. M. Fernández-Caramés, "Green IoT and edge AI as key technological enablers for a sustainable digital transition towards a smart circular economy: An industry 5.0 use case," *Sensors*, vol. 21, no. 17, p. 5745, 2021.

[231] O. A. Amodu, U. A. Bukar, R. A. R. Mahmood, C. Jarray, and M. Othman, "Age of information minimization in UAV-aided data collection for WSN and IoT applications: A systematic review," *J. Netw. Comput. Appl.*, vol. 216, Jul. 2023, Art. no. 103652.

[232] A. Mahmood et al., "Industrial IoT in 5G-and-beyond networks: Vision, architecture, and design trends," *IEEE Trans. Ind. Informat.*, vol. 18, no. 6, pp. 4122–4137, Jun. 2022.

[233] O. Elijah, S. K. Abdul Rahim, W. K. New, C. Y. Leow, K. Cumanan, and T. Kim Geok, "Intelligent massive MIMO systems for beyond 5G networks: An overview and future trends," *IEEE Access*, vol. 10, pp. 102532–102563, 2022.

[234] R. Liu, G. Yu, J. Yuan, and G. Y. Li, "Resource management for millimeter-wave ultra-reliable and low-latency communications," *IEEE Trans. Commun.*, vol. 69, no. 2, pp. 1094–1108, Feb. 2021.

[235] I. F. Akyildiz, A. Kak, and S. Nie, "6G and beyond: The future of wireless communications systems," *IEEE Access*, vol. 8, pp. 133995–134030, 2020.

[236] C.-X. Wang et al., "On the road to 6G: Visions, requirements, key technologies, and testbeds," *IEEE Commun. Surveys Tuts.*, vol. 25, no. 2, pp. 905–974, 2nd Quart., 2023.

[237] T. Gong, L. Zhu, F. R. Yu, and T. Tang, "Edge intelligence in intelligent transportation systems: A survey," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 9, pp. 8919–8944, Sep. 2023.

[238] M. Hijji et al., "6G connected vehicle framework to support intelligent road maintenance using deep learning data fusion," *IEEE Trans. Intell. Transp. Syst.*, vol. 24, no. 7, pp. 7726–7735, Jul. 2023.

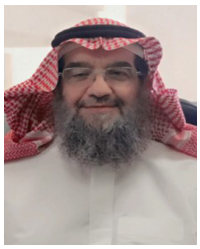
[239] L. U. Khan, Y. K. Tun, M. Alsenwi, M. Imran, Z. Han, and C. S. Hong, "A dispersed federated learning framework for 6G-enabled autonomous driving cars," *IEEE Trans. Netw. Sci. Eng.*, vol. 11, no. 6, pp. 5656–5667, Nov./Dec. 2024.

[240] Y. Li, "Deep reinforcement learning: An overview," 2017, *arXiv:1701.07274*.

[241] J. Zheng, K. Li, N. Mhaisen, W. Ni, E. Tova, and M. Guizani, "Exploring deep-reinforcement-learning-assisted federated learning for online resource allocation in privacy-preserving edgeIoT," *IEEE Internet Things J.*, vol. 9, no. 21, pp. 21099–21110, Nov. 2022.
[242] J. Chen, J. Li, R. Huang, K. Yue, Z. Chen, and W. Li, "Federated transfer learning for bearing fault diagnosis with discrepancy-based weighted federated averaging," *IEEE Trans. Instrum. Meas.*, vol. 71, pp. 1–11, 2022.
[243] W. Yang, W. Xiang, Y. Yang, and P. Cheng, "Optimizing federated learning with deep reinforcement learning for digital twin empowered Industrial IoT," *IEEE Trans. Ind. Informat.*, vol. 19, no. 2, pp. 1884–1893, Feb. 2023.
[244] P. Ladosz, L. Weng, M. Kim, and H. Oh, "Exploration in deep reinforcement learning: A survey," *Inf. Fusion*, vol. 85, pp. 1–22, Sep. 2022.
[245] Y. L. Lee and D. Qin, "A survey on applications of deep reinforcement learning in resource management for 5G heterogeneous networks," in *Proc. Asia–Pacific Signal Inf. Process. Assoc. Annu. Summit Conf. (APSIPA ASC)*, 2019, pp. 1856–1862.
[246] Y. Chen, Z. Liu, Y. Zhang, Y. Wu, X. Chen, and L. Zhao. "Deep reinforcement learning-based dynamic resource management for mobile edge computing in Industrial Internet of Things," *IEEE Trans. Ind. Informat.*, vol. 17, no. 7, pp. 4925–4934, Jul. 2021.
[247] P. Tam, S. Math, C. Nam, and S. Kim, "Adaptive resource optimized edge federated learning in real-time image sensing classifications," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 14, pp. 10929–10940, 2021.
[248] J. Choi and S. R. Pokhrel, "Federated learning with multichannel aloha," *IEEE Wireless Commun. Lett.*, vol. 9, no. 4, pp. 499–502, Apr. 2020.
[249] G. Rjoub, O. A. Wahab, J. Bentahar, R. Cohen, and A. S. Bataineh, "Trust-augmented deep reinforcement learning for federated learning client selection," *Inf. Syst. Front.*, vol. 26, pp. 1261–1278, Aug. 2024.
[250] Y. Guo, Z. Zhao, K. He, S. Lai, J. Xia, and L. Fan, "Efficient and flexible management for Industrial Internet of Things: A federated learning approach," *Comput. Netw.*, vol. 192, Jun. 2021, Art. no. 108122.
[251] T. Ryffel et al., "A generic framework for privacy preserving deep learning," 2018, *arXiv:1811.04017*.
[252] "Tensorflow federated," TensorFlow, 2016. Accessed: Dec. 15, 2023. [Online]. Available: https://www.tensorflow.org/federated
[253] S. Caldas et al., "Leaf: A benchmark for federated settings," 2018, *arXiv:1812.01097*.
[254] C. He et al., "FedML: A research library and benchmark for federated machine learning," 2020, *arXiv:2007.13518*.
[255] *An Industrial Grade Federated Learning Framework*, WeBank, Shenzhen, China, Dec. 2023.
[256] D. J. Beutel et al., "Flower: A friendly federated learning research framework," 2020, *arXiv:2007.14390*.
[257] S. Das Sarma, D. L. Deng, and L. M. Duan, "Machine learning meets quantum physics," *Phys. Today*, vol. 72, no. 3, pp. 48–54, Mar. 2019.
[258] H. T. Larasati, M. Firdaus, and H. Kim, "Quantum federated learning: Remarks and challenges," in *Proc. IEEE 9th Int. Conf. Cyber Security Cloud Comput. (CSCloud), IEEE 8th Int. Conf. Edge Comput. Scalable Cloud (EdgeCom)*, Jun. 2022, pp. 1–5.
[259] S. Y. C. Chen and S. Yoo, "Federated quantum machine learning," *Entropy*, vol. 23, no. 4, p. 460, Apr. 2021.



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